

## Stability Analysis

- stability: stable working condition of a control system.
- In a stable system, the output is predictable, finite and stable for a given input.
- If output is bounded (finite) for any bounded input  $\Rightarrow$  system is stable.
- If output tends towards zero irrespective of initial conditions or in the absence of the input  $\Rightarrow$  system is asymptotically stable.
- If output is of infinite amplitude or oscillatory for a bounded disturbing input signal  $\Rightarrow$  system is unstable.
- If output has constant amplitude oscillations for a bounded input signal  $\Rightarrow$  system maybe stable or unstable under some limited constraints. [Limitedly stable]
- If output is stable for all variations of its parameters  $\Rightarrow$  system is absolutely stable.
- If output is stable for a limited range of variations of its parameters  $\Rightarrow$  system is conditionally stable.

## Location of Roots

$$T(s) = \frac{C(s)}{R(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} = \frac{(s+z_1) \dots (s+z_m)}{(s+p_1) \dots (s+p_n)} = \frac{A_1}{s+p_1} + \dots + \frac{A_n}{s+p_n}$$

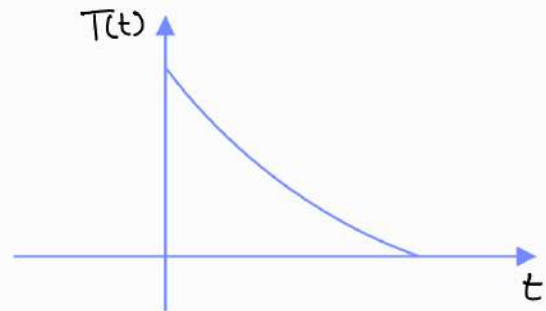
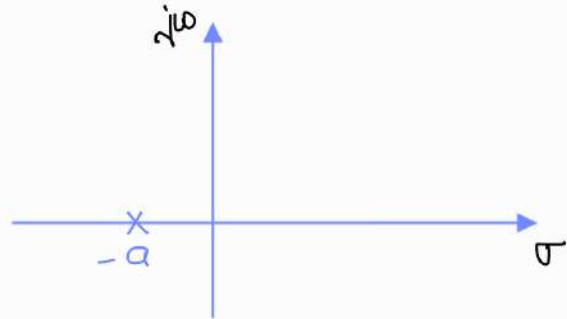
→ roots on negative real axis:

$$T(s) = \frac{A}{s+a}$$

impulse response:

$$T(t) = A e^{-at}$$

response is exponentially decaying  $\Rightarrow$  stable



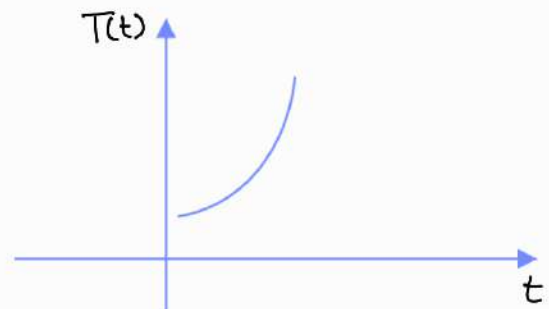
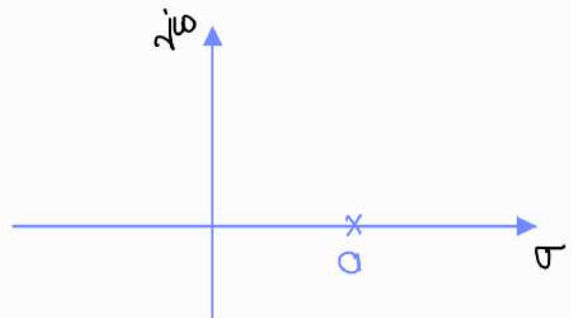
→ roots on negative real axis:

$$T(s) = \frac{A}{s-a}$$

impulse response:

$$T(t) = A e^{at}$$

response is exponentially increasing  $\Rightarrow$  unstable

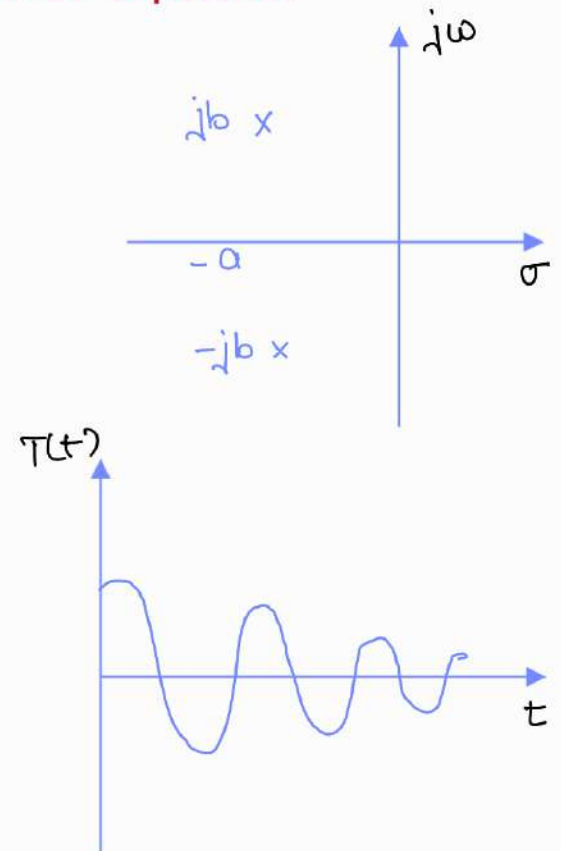


→ complex conjugate roots on left half of s-plane:

$$T(s) = \frac{A}{s+a+jb} + \frac{A}{s+a-jb}$$

impulse response:

$$\begin{aligned} T(t) &= A e^{-(a+jb)t} + A e^{-(a-jb)t} \\ &= 2A e^{-at} \cos bt \\ &= 2A e^{-at} \sin(bt + 90^\circ) \end{aligned}$$



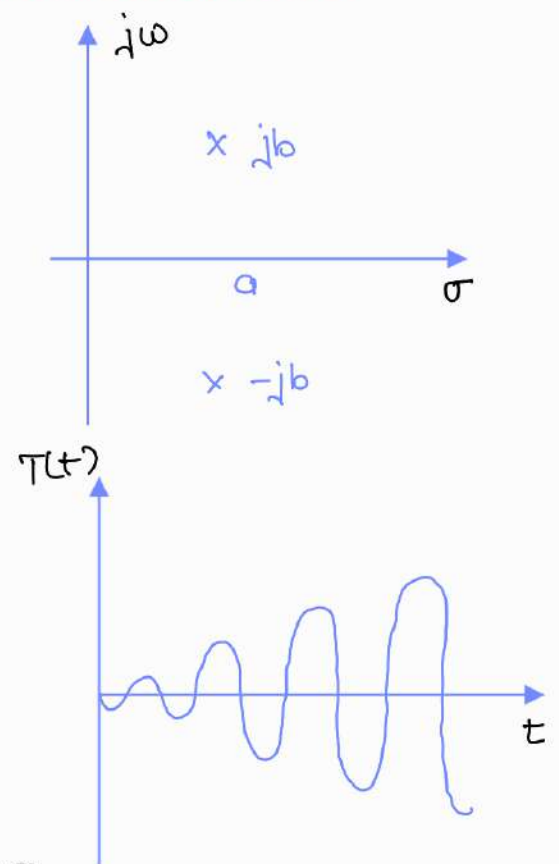
response is damped sinusoidal/oscillatory ⇒ stable.

→ complex conjugate roots on right half of s-plane:

$$T(s) = \frac{A}{s-a+jb} + \frac{A}{s-a-jb}$$

impulse response:

$$\begin{aligned} T(t) &= A e^{(a+jb)t} + A e^{(a-jb)t} \\ &= 2A e^{at} \cos bt \\ &= 2A e^{at} \sin(bt + 90^\circ) \end{aligned}$$



response is exponentially increasing sinusoidal ⇒ unstable.

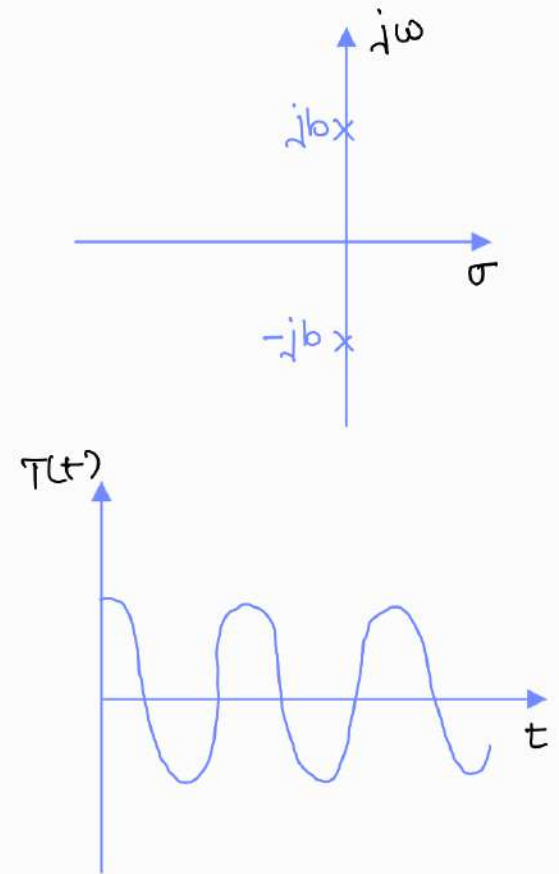
→ single pair of roots on imaginary axis:

$$T(s) = \frac{A}{s+jb} + \frac{A}{s-jb}$$

imaginary axis:

$$\begin{aligned} T(t) &= A e^{-jbt} + A e^{jbt} \\ &= 2A \cos bt \\ &= 2A \sin(bt + 90^\circ) \end{aligned}$$

response is oscillatory  $\Rightarrow$   
marginally stable



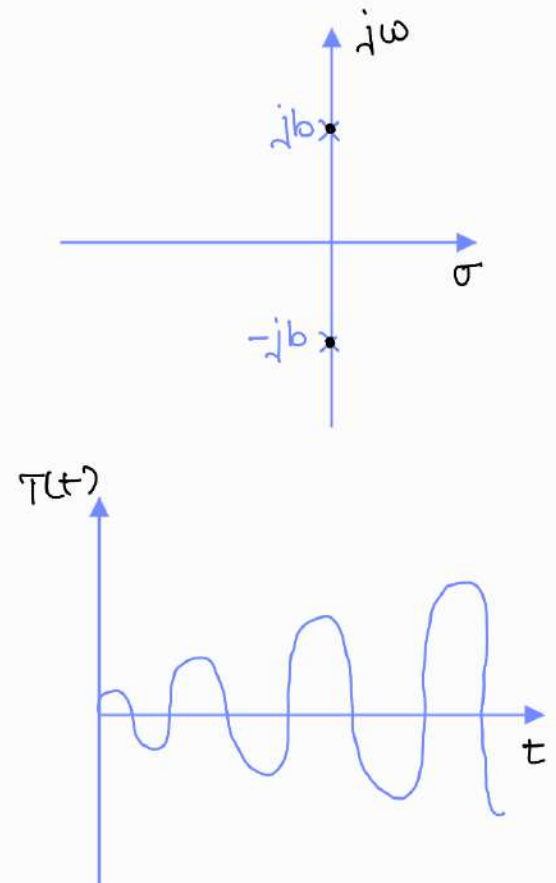
→ double pair of roots on imaginary axis:

$$T(s) = \frac{A}{(s+jb)^2} + \frac{A}{(s-jb)^2}$$

imaginary axis:

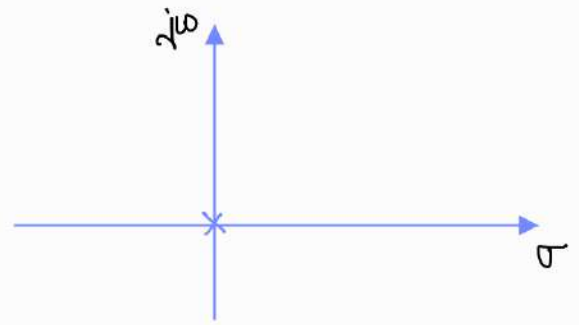
$$\begin{aligned} T(t) &= At e^{-jbt} + At e^{jbt} \\ &= 2At \cos bt \\ &= 2A t \sin(bt + 90^\circ) \end{aligned}$$

response is linearly increasing  
sinusoidal  $\Rightarrow$  unstable



→ single root at origin:

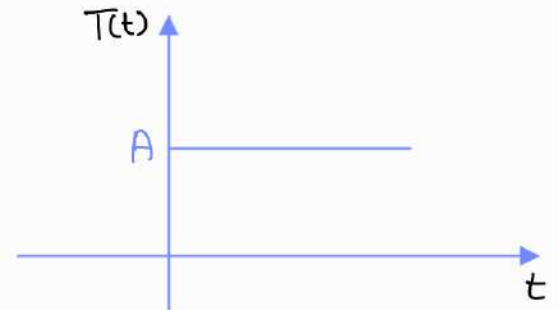
$$T(s) = \frac{A}{s}$$



impulse response:

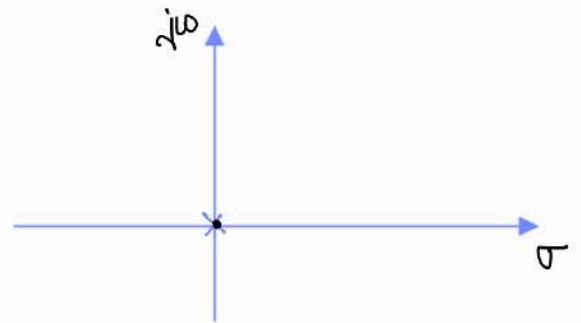
$$T(t) = A$$

response is constant  $\Rightarrow$   
marginally stable



→ double root at origin:

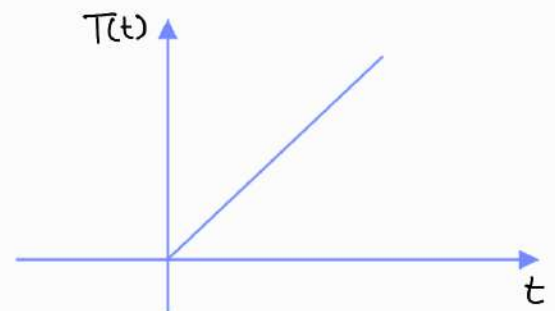
$$T(s) = \frac{A}{s^2}$$



impulse response:

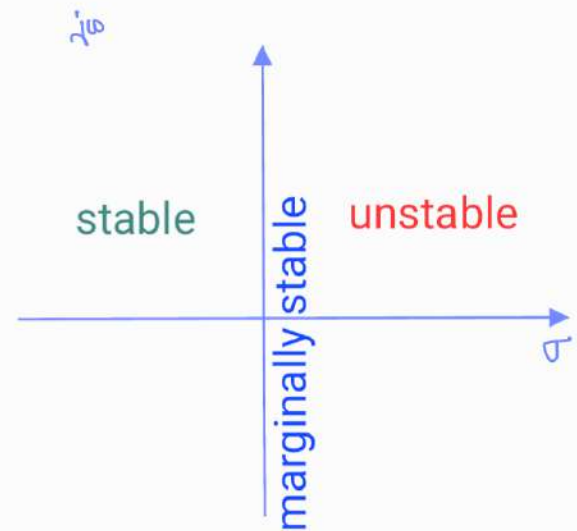
$$T(t) = At$$

response is linearly increase  
with time  $\Rightarrow$  unstable



## stability of the system depending on location of roots:

- If all the roots of characteristic equation has negative real parts  $\Rightarrow$  system is stable.
- If any root of the characteristic equation has a positive real part or if there is a repeated root on the imaginary axis  $\Rightarrow$  system is unstable.
- If all the root of characteristic equation has negative real parts except for the presence of one or more non repeated roots on the imaginary axis  $\Rightarrow$  system is limitedly or marginally stable.



## Coefficients of characteristic polynomial:

$$a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_{n-1} s^1 + a_n = 0$$

$r_1, r_2, r_3, \dots, r_n$  roots are functions of coefficients  $a_0, a_1, \dots, a_n$

- If all the coefficients are positive and if no coefficient is zero  $\Rightarrow$  all the roots are in the left half of s-plane.
- If any Coefficient is equal to zero  $\Rightarrow$  some of the roots may be on the imaginary axis or on the right half of s-plane.
- If any coefficient is negative  $\Rightarrow$  atleast one root is in the right half of s-plane.

## Routh-Hurwitz stability criterion

→ analytical procedure for determining whether all the roots of a polynomial have negative real part or not.

→ Routh criterion ⇒ array formulation

→ Hurwitz criterion ⇒ determinants

Characteristic equation:

$$a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_{n-1} s^1 + a_n = 0 \quad ; \quad a_0 > 0$$

Routh array:

|             |                                                                                                 |       |       |         |
|-------------|-------------------------------------------------------------------------------------------------|-------|-------|---------|
| $s^n :$     | $a_0$                                                                                           | $a_2$ | $a_4$ | $\dots$ |
| $s^{n-1} :$ | $a_1$                                                                                           | $a_3$ | $a_5$ | $\dots$ |
|             |                                                                                                 |       |       |         |
| $s^{n-2} :$ | $b_0 = \frac{a_1 a_2 - a_0 a_3}{a_1} \quad b_1 = \frac{a_1 a_4 - a_0 a_5}{a_1} \quad b_2 \dots$ |       |       |         |
| $s^{n-3} :$ | $c_0 = \frac{b_0 a_3 - a_1 b_1}{b_0} \quad c_1 = \frac{b_0 a_5 - a_1 b_2}{b_0} \quad c_2 \dots$ |       |       |         |
| $s^1 :$     | $g_0$                                                                                           |       |       |         |
| $s^0 :$     | $h_0$                                                                                           |       |       |         |

Hurwitz determinants:

$$b_0 = - \frac{\begin{vmatrix} a_0 & a_2 \\ a_1 & a_3 \end{vmatrix}}{a_1} \quad ; \quad c_0 = - \frac{\begin{vmatrix} a_1 & a_3 \\ b_0 & b_1 \end{vmatrix}}{b_0}$$

→ stability criterion: all of the elements in the first column of the Routh array be positive

→ number of sign changes ( $+$   $\rightarrow$   $-$  or  $-$   $\rightarrow$   $+$ )  $\Rightarrow$  number of roots of the characteristic equation in the right half of the s-plane.

→ cases:

↳ non-zero elements in the first column

↳ row of all zeroes

↳ first element of row is zero but some or other elements are not zero

### Normal Routh array:

→ non-zero elements in first column of Routh array.

→ If there is no sign change in the first column of Routh array  $\Rightarrow$  all the roots are lying on left half of s-plane  $\Rightarrow$  system is stable

→ If there is sign change in the first column of Routh array  $\Rightarrow$  number of sign changes is equal to number of roots lying on the right half of s-plane  $\Rightarrow$  system is unstable

1. Using Routh criterion, determine the stability of the system represented by the following characteristic equation. Comment on the location of the roots of characteristic equation.

$$s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$$

|       | $1s^4 + 8s^3 + 18s^2 + 16s + 5s^0$         |                                     |   |
|-------|--------------------------------------------|-------------------------------------|---|
| $s^4$ | 1                                          | 18                                  | 5 |
| $s^3$ | 8                                          | 16                                  |   |
|       | $(\div 8)$                                 |                                     |   |
| $s^3$ | 1                                          | 2                                   |   |
| $s^2$ | $\frac{1 \times 18 - 1 \times 2}{1}$       | $\frac{1 \times 5 - 1 \times 0}{1}$ |   |
| $s^2$ | 16                                         | 5                                   |   |
| $s^1$ | $\frac{16 \times 2 - 1 \times 5}{16}$      |                                     |   |
| $s^1$ | 1.69                                       |                                     |   |
| $s^0$ | $\frac{1.69 \times 5 - 16 \times 0}{1.69}$ |                                     |   |
| $s^0$ | 5                                          |                                     |   |

→ all elements of first column non-zero and there is no sign change ⇒ roots are lying on the left half of s-plane  
⇒ system is stable.

2. By Routh stability criterion determine the stability of the system represented by the following characteristic equation. comment on the location of roots of characteristic equation.

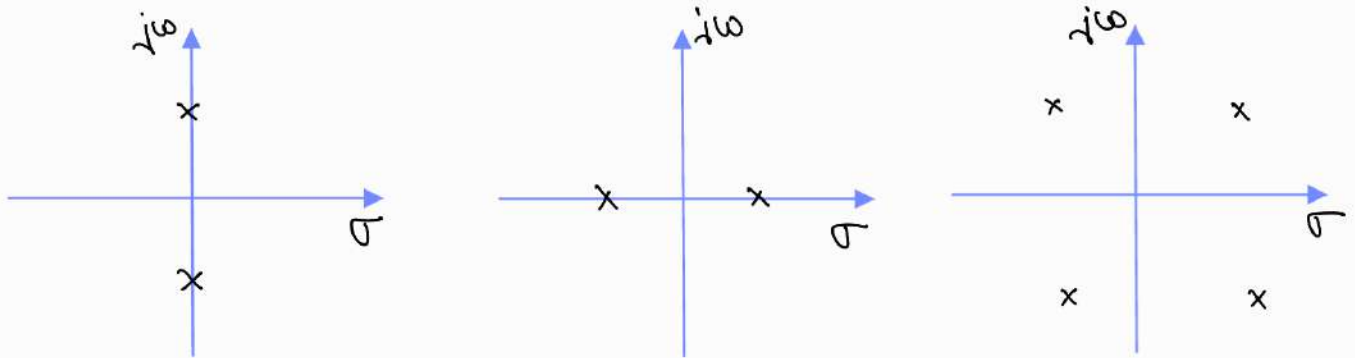
$$9s^5 - 20s^4 + 10s^3 - s^2 - 9s - 10 = 0$$

|                         |                                                        |                                                 |     |
|-------------------------|--------------------------------------------------------|-------------------------------------------------|-----|
|                         | $9s^5 - 20s^4 + 10s^3 - s^2 - 9s - 10$                 |                                                 |     |
|                         | <hr/>                                                  |                                                 |     |
| $s^5$                   | 9                                                      | 10                                              | -9  |
| $+ \rightarrow -$ $s^4$ | -20                                                    | -1                                              | -10 |
| $s^3$                   | $\frac{-20 \times 10 - (9 \times -1)}{-20}$            | $\frac{-20 \times -9 - (9 \times -10)}{-20}$    |     |
| $- \rightarrow +$ $s^3$ | 9.55                                                   | -13.5                                           |     |
| $s^2$                   | $\frac{9.55 \times -1 - (-20 \times -13.5)}{9.55}$     | $\frac{9.55 \times -10 - (-20 \times 0)}{9.55}$ |     |
| $+ \rightarrow -$ $s^2$ | -29.3                                                  | -10                                             |     |
| $s^1$                   | $\frac{-29.3 \times -13.5 - (9.55 \times -10)}{-29.3}$ |                                                 |     |
| $s^1$                   | -16.8                                                  |                                                 |     |
| $s^0$                   | $\frac{-29.3 \times 0 - (-10 \times -16.8)}{-16.8}$    |                                                 |     |
| $s^0$                   | -10                                                    |                                                 |     |

$\rightarrow$  all the elements are non-zero and there are three sign changes  $\Rightarrow$  three roots are lying in right half of s-plane and two roots are lying in left half of s-plane  $\Rightarrow$  system is unstable.

## Auxiliary Polynomial:

- a row of all zeros.
- In even polynomial  $\Rightarrow$  exponents of  $s$  are even integers or zero.
- Coefficients of auxiliary polynomial  $\Rightarrow$  elements of row directly above the zeros in the array.
- roots are purely imaginary, purely real or complex



- ↳ determine the auxiliary polynomial  $A(s)$ .
- ↳ differentiate the  $A(s)$  with respect to  $s \Rightarrow dA(s)/ds$
- ↳ replace the row of zeros with coefficients of  $dA(s)/ds$ .
- ↳ If there are sign changes in the first column  $\Rightarrow$  number of sign changes is equal to number of roots lying on right half of  $s$ -plane  $\Rightarrow$  system is unstable
- ↳ If there are no sign changes in the first column  $\Rightarrow$  zeros row indicate the existence of purely imaginary roots  $\Rightarrow$  system is marginally stable.
- ↳ Roots of auxiliary equation lies on imaginary axis and the remaining roots are lying on the left half of  $s$ -plane.

3. Construct Routh array and determine the stability of the system whose characteristic equation is given below. Also determine The number of roots lying on right half of s-plane, left half of s-plane and on imaginary axis.

$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$

|       | $1s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s^1 + 16s^0$ |                                      |                                       |    |
|-------|------------------------------------------------------|--------------------------------------|---------------------------------------|----|
| $s^6$ | 1                                                    | 8                                    | 20                                    | 16 |
| $s^5$ | 2                                                    | 12                                   | 16                                    |    |
|       |                                                      | $(\div 2)$                           |                                       |    |
| $s^5$ | 1                                                    | 6                                    | 8                                     |    |
| $s^4$ | $\frac{1 \times 8 - 1 \times 6}{1}$                  | $\frac{1 \times 20 - 1 \times 8}{1}$ | $\frac{1 \times 16 - 1 \times 0}{1}$  |    |
| $s^4$ | 2                                                    | 12                                   | 16                                    |    |
|       |                                                      | $(\div 2)$                           |                                       |    |
| $s^4$ | 1                                                    | 6                                    |                                       |    |
|       |                                                      |                                      | $8 \Rightarrow A(s) = s^4 + 6s^2 + 8$ |    |
| $s^3$ | $\frac{1 \times 6 - 1 \times 6}{1}$                  | $\frac{1 \times 8 - 1 \times 8}{1}$  | $\frac{dA(s)}{ds} = 4s^3 + 12s$       |    |
| $s^3$ | 0                                                    | 0                                    |                                       |    |
| $s^3$ | 4                                                    | 12                                   |                                       |    |
|       |                                                      | $(\div 4)$                           |                                       |    |
| $s^3$ | 1                                                    | 3                                    |                                       |    |
| $s^2$ | $\frac{1 \times 6 - 1 \times 3}{1}$                  | $\frac{1 \times 8 - 1 \times 0}{1}$  |                                       |    |
| $s^2$ | 3                                                    | 8                                    |                                       |    |
| $s^1$ | $\frac{3 \times 3 - 1 \times 8}{3}$                  |                                      |                                       |    |
| $s^1$ | 0.33                                                 |                                      |                                       |    |
| $s^0$ | $\frac{0.33 \times 8 - 3 \times 0}{0.33}$            | 8                                    |                                       |    |

→ there are no sign changes in the first column ⇒ zeros row indicate the existence of purely imaginary roots ⇒ system is marginally stable.

$$s^4 + 6s^2 + 8 = 0$$

$$\text{let } x = s^2$$

$$\Rightarrow x^2 + 6x + 8 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-6 \pm \sqrt{36 - 32}}{2}$$

$$x = -3 \pm 1$$

$$x = -2 \text{ or } -4$$

$$x = s^2$$

$$s = \sqrt{x}$$

$$s = \sqrt{-2} \quad \Bigg| \quad s = \sqrt{-4}$$

$$s = \pm \sqrt{2}j \quad \Bigg| \quad s = \pm 2j$$

→ four roots of auxiliary equation lies on imaginary axis and the remaining two roots are lying on the left half of s-plane.

4. Construct Routh array and determine the stability of the system whose characteristic equation is given below. Also determine The number of roots lying on right half of s-plane, left half of s-plane and on imaginary axis.

$$s^6 + s^5 + 5s^4 + 3s^3 + 2s^2 - 4s - 8 = 0$$

|       | $s^6 + s^5 + 5s^4 + 3s^3 + 2s^2 - 4s - 8$        |                                         |                                      |                                        |
|-------|--------------------------------------------------|-----------------------------------------|--------------------------------------|----------------------------------------|
| $s^6$ | 1                                                | 5                                       | 2                                    | -8                                     |
| $s^5$ | 1                                                | 3                                       | -4                                   |                                        |
| $s^4$ | $\frac{1 \times 5 - 1 \times 3}{1}$              | $\frac{1 \times 2 - (1 \times -4)}{1}$  | $\frac{1 \times -8 - 1 \times 0}{1}$ |                                        |
| $s^4$ | 2                                                | 6<br>( $\div 2$ )                       | -8                                   |                                        |
| $s^4$ | 1                                                | 3                                       |                                      | $-4 \Rightarrow A(s) = s^4 + 3s^2 - 4$ |
| $s^3$ | $\frac{1 \times 3 - 1 \times 3}{1}$              | $\frac{1 \times -4 - (1 \times -4)}{1}$ |                                      | $\frac{dA(s)}{ds} = 4s^3 + 6s$         |
| $s^3$ | 0                                                | 0                                       |                                      |                                        |
| $s^3$ | 2<br>( $\div 2$ )                                | 6                                       |                                      |                                        |
| $s^3$ | 2                                                | 3                                       |                                      |                                        |
| $s^2$ | $\frac{2 \times 3 - 1 \times 3}{2}$              | $\frac{2 \times -4 - 1 \times 0}{2}$    |                                      |                                        |
| $s^2$ | 1.5                                              | -4                                      |                                      |                                        |
| $s^1$ | $\frac{1.5 \times 3 - (2 \times -4)}{1.5}$       |                                         |                                      |                                        |
| $s^1$ | 8.33                                             |                                         |                                      |                                        |
| $s^0$ | $\frac{(8.33 \times -4) - (1.5 \times 0)}{8.33}$ |                                         |                                      |                                        |
| $s^0$ | -4                                               |                                         |                                      |                                        |

$\rightarrow -$

→ there is a sign change in the first column  $\Rightarrow$  one root is lying on right half of s-plane  $\Rightarrow$  system is unstable

$$s^4 + 3s^2 - 4 = 0$$

let  $x = s^2$

$$x^2 + 3x - 4 = 0$$

$$(x+4)(x-1) = 0$$

$$x = -4, 1$$

|                 |                |
|-----------------|----------------|
| $x = -4$        | $x = 1$        |
| $s^2 = -4$      | $s^2 = 1$      |
| $s = \sqrt{-4}$ | $s = \sqrt{1}$ |
| $s = \pm 2j$    | $s = \pm 1$    |

$$s^6 + s^5 + 5s^4 + 3s^3 + 2s^2 - 4s - 8 = 0$$

$$(s^4 + 3s^2 - 4)(s^2 + s + 2) = 0$$

$$s^2 + s + 2 = 0$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1 - 4 \times 2}}{2}$$

$$= \frac{-1 \pm \sqrt{-7}}{2}$$

$$s = \frac{-1 \pm \sqrt{7}j}{2}$$

→ two roots of auxiliary equation lies on imaginary axis and the remaining three roots are lying on the left half of s-plane.

5. Construct Routh array and determine the stability of the system whose characteristic equation is given below. Also determine The number of roots lying on right half of s-plane, left half of s-plane and on imaginary axis.

$$s^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2 = 0$$

|       | $1s^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s^1 + 2s^0$ |                                     |                                     |   |
|-------|--------------------------------------------------|-------------------------------------|-------------------------------------|---|
| $s^6$ | 1                                                | 4                                   | 5                                   | 2 |
| $s^5$ | 3                                                | 6<br>( $\div 3$ )                   | 3                                   |   |
| $s^4$ | 1                                                | 2                                   | 1                                   |   |
| $s^4$ | $\frac{1 \times 4 - 1 \times 2}{1}$              | $\frac{1 \times 5 - 1 \times 1}{1}$ | $\frac{1 \times 2 - 1 \times 0}{1}$ |   |
| $s^4$ | 2                                                | 4<br>( $\div 2$ )                   | 2                                   |   |
| $s^4$ | 1                                                | 2                                   |                                     |   |
| $s^3$ | $\frac{1 \times 2 - 2 \times 1}{1}$              | $\frac{1 \times 1 - 1 \times 1}{1}$ |                                     |   |
| $s^3$ | 0                                                | 0                                   |                                     |   |
| $s^3$ | 4<br>( $\div 4$ )                                | 4                                   |                                     |   |
| $s^3$ | 1                                                | 1                                   |                                     |   |
| $s^2$ | $\frac{1 \times 2 - 1 \times 1}{1}$              | $\frac{1 \times 1 - 1 \times 0}{1}$ |                                     |   |
| $s^2$ | 1                                                | 1                                   |                                     |   |
| $s^1$ | $\frac{1 \times 1 - 1 \times 1}{1}$              |                                     |                                     |   |
| $s^1$ | 0                                                |                                     |                                     |   |
| $s^1$ | 2                                                |                                     |                                     |   |
| $s^0$ | $\frac{2 \times 1 - 1 \times 0}{2} = 1$          |                                     |                                     |   |

$$1 \Rightarrow A(s) = s^4 + 2s^2 + 1$$

$$\frac{dA(s)}{ds} = 4s^3 + 4s$$

$$\Rightarrow B(s) = s^2 + 1$$

$$\frac{dB(s)}{ds} = 2s$$

→ there are no sign changes in the first column ⇒ zeros row indicate the existence of purely imaginary roots ⇒ system is marginally stable.

$$s^4 + 2s^2 + 1 = 0$$

$$\text{Let } x = s^2$$

$$x^2 + 2x + 1 = 0$$

$$(x+1)^2 = 0$$

$$x = -1, -1$$

$$x = s^2$$

$$\begin{array}{l|l} x = -1 \Rightarrow s = \sqrt{x} & x = -1 \Rightarrow s = \sqrt{-1} \\ s = \sqrt{-1} & s = \sqrt{-1} \\ s = \pm j & s = \pm j \end{array}$$

$$s^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2 = 0$$

$$(s^4 + 2s^2 + 1)(s^2 + 3s + 2) = 0$$

$$s^2 + 3s + 2 = 0$$

$$(s+2)(s+1) = 0$$

$$s = -1, -2$$

→ four roots of auxiliary equation lies on imaginary axis and the remaining two roots are lying on the left half of s-plane.

## First element of a row is zero but some or other elements are not zero

→ If a zero is encountered as first element of a row  $\Rightarrow$  all the elements of the next row will be infinite.

↳ Let  $0 \rightarrow \xi$  and complete the array then  $\xi \rightarrow 0$  and determine the values of elements of array which are function of  $\xi$

↳ If there is no sign change in first column of Routh array and if there is no row with all zeros  $\Rightarrow$  all the roots are lying on left half of s-plane  $\Rightarrow$  system is stable.

↳ If there are sign changes in first column of Routh array and there is no row with all zeros  $\Rightarrow$  number of sign changes is equal to number of roots lying on right half of s-plane  $\Rightarrow$  system is unstable.

↳ If there is a row of all zeros after letting  $\xi \rightarrow 0 \Rightarrow$  zeros row indicate the existence of purely imaginary roots  $\Rightarrow$  system is marginally stable.

↳ Roots of auxiliary equation lies on imaginary axis and the remaining roots are lying on the left half of s-plane.

6. Construct Routh array and determine the stability of the system represented by the following characteristic equation. Comment on the location of the roots of characteristic equation.

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

|       | $1s^5 + 1s^4 + 2s^3 + 2s^2 + 3s^1 + 5s^0$                                           |                           |   |
|-------|-------------------------------------------------------------------------------------|---------------------------|---|
| $s^5$ | 1                                                                                   | 2                         | 3 |
| $s^4$ | 1                                                                                   | 2                         | 5 |
| $s^3$ | $\frac{1*2 - 1*2}{1}$                                                               | $\frac{1*3 - 1*5}{1}$     |   |
| $s^3$ | 0 $\rightarrow \xi$                                                                 | -2                        |   |
| $s^3$ | $\xi$                                                                               | -2                        |   |
| $s^2$ | $\frac{\xi*2 - (1*-2)}{\xi}$                                                        | $\frac{\xi*5 - 1*0}{\xi}$ |   |
| $s^2$ | $\frac{2\xi + 2}{\xi}$                                                              | 5                         |   |
| $s^1$ | $\frac{2\xi + 2 * -2 - \xi * 5}{\xi}$                                               |                           |   |
| $s^1$ | $\frac{2\xi + 2}{\xi}$                                                              |                           |   |
| $s^1$ | $-\frac{(5\xi^2 + 4\xi + 4)}{2\xi + 2}$                                             |                           |   |
| $s^0$ | $\frac{-(5\xi^2 + 4\xi + 4) * 5 - \left(\frac{2\xi + 2}{\xi}\right) * 0}{2\xi + 2}$ |                           |   |
| $s^0$ | $\frac{-(5\xi^2 + 4\xi + 4)}{2\xi + 2}$                                             |                           |   |
| $s^0$ | 5                                                                                   |                           |   |

letting  $\xi \rightarrow 0$

|       |          |    |   |
|-------|----------|----|---|
| $s^5$ | 1        | 2  | 3 |
| $s^4$ | 1        | 2  | 5 |
| $s^3$ | 0        | -2 |   |
| $s^2$ | $\infty$ | 5  |   |
| $s^1$ | -2       |    |   |
| $s^0$ | 5        |    |   |

$+\rightarrow -$

$-\rightarrow +$

$\rightarrow$  there are two sign changes in first column of Routh array and there is no row with all zeros  $\Rightarrow$  two roots lying on right half of s-plane and the remaining three roots are lying on the left half of s-plane.  $\Rightarrow$  system is unstable.

7. Use the Routh stability criterion to determine the location of roots on the s-plane and hence the stability for the system represented by the characteristic equation.

$$s^5 + 4s^4 + 8s^3 + 8s^2 + 7s + 4 = 0$$

|       | $s^5 + 4s^4 + 8s^3 + 8s^2 + 7s + 4$ |                                     |   |
|-------|-------------------------------------|-------------------------------------|---|
| $s^5$ | 1                                   | 8                                   | 7 |
| $s^4$ | 4                                   | 8<br>( $\div 4$ )                   | 4 |
| $s^3$ | 1                                   | 2                                   | 1 |
| $s^2$ | $\frac{1 \times 8 - 1 \times 2}{1}$ | $\frac{1 \times 7 - 1 \times 1}{1}$ |   |
| $s^1$ | 6                                   | 6                                   |   |
| $s^0$ | 4                                   |                                     |   |

|       |                                         |                                     |
|-------|-----------------------------------------|-------------------------------------|
| $s^2$ | $\frac{1 \times 2 - 1 \times 1}{1}$     | $\frac{1 \times 1 - 1 \times 0}{1}$ |
| $s^2$ | 1                                       | 1                                   |
| $s^1$ | $\frac{1 \times 1 - 1 \times 1}{1}$     |                                     |
| $s^1$ | 0 $\rightarrow \xi$                     |                                     |
| $s^1$ | $\xi$                                   |                                     |
| $s^0$ | $\frac{1 \times \xi - 1 \times 0}{\xi}$ |                                     |
| $s^0$ | 1                                       |                                     |

Letting  $\xi \rightarrow 0$

|       |   |   |   |
|-------|---|---|---|
| $s^5$ | 1 | 8 | 7 |
| $s^4$ | 1 | 2 | 1 |
| $s^3$ | 1 | 1 |   |
| $s^2$ | 1 | 1 |   |
| $s^1$ | 0 |   |   |
| $s^0$ | 1 |   |   |

$\rightarrow$  there is a row of all zeros after letting  $\xi \rightarrow 0 \Rightarrow$  zeros row indicate the existence of purely imaginary roots  $\Rightarrow$  system is marginally stable.

$$A(s) = s^2 + 1 = 0$$

$$s^2 = -1$$

$$s = \pm j$$

$\rightarrow$  two roots of auxiliary equation lies on imaginary axis and three roots are lying on the left half of s-plane.

8. Determine the range of  $k$  for stability of unity feedback system whose open loop transfer function is  $G(s)$ .

$$G(s) = \frac{k}{s(s+1)(s+2)}$$

Transfer function:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)}$$

unity feedback:

$$H(s) = 1$$

$$T(s) = \frac{\frac{k}{s(s+1)(s+2)}}{1 + \frac{k}{s(s+1)(s+2)} * 1}$$

$$T(s) = \frac{k}{s(s+1)(s+2) + k}$$

characteristic equation:

$$s(s+1)(s+2) + k = 0$$

$$s(s^2 + 3s + 2) + k = 0$$

$$s^3 + 3s^2 + 2s + k = 0$$

|       | $1s^3 + 3s^2 + 2s^1 + ks^0$ |
|-------|-----------------------------|
| $s^3$ | 1                      2    |
| $s^2$ | 3                      k    |
| $s^1$ | $\frac{3*2 - 1*k}{3}$       |
| $s^0$ | $\frac{6-k}{3}$             |

$$\begin{array}{c|c}
 s^0 & \frac{6-k}{3} \neq 0 \\
 \hline
 & \frac{6-k}{3} \\
 s^1 & k
 \end{array}$$

→ from  $s^0$  row: for the system to be stable  $\Rightarrow k > 0$

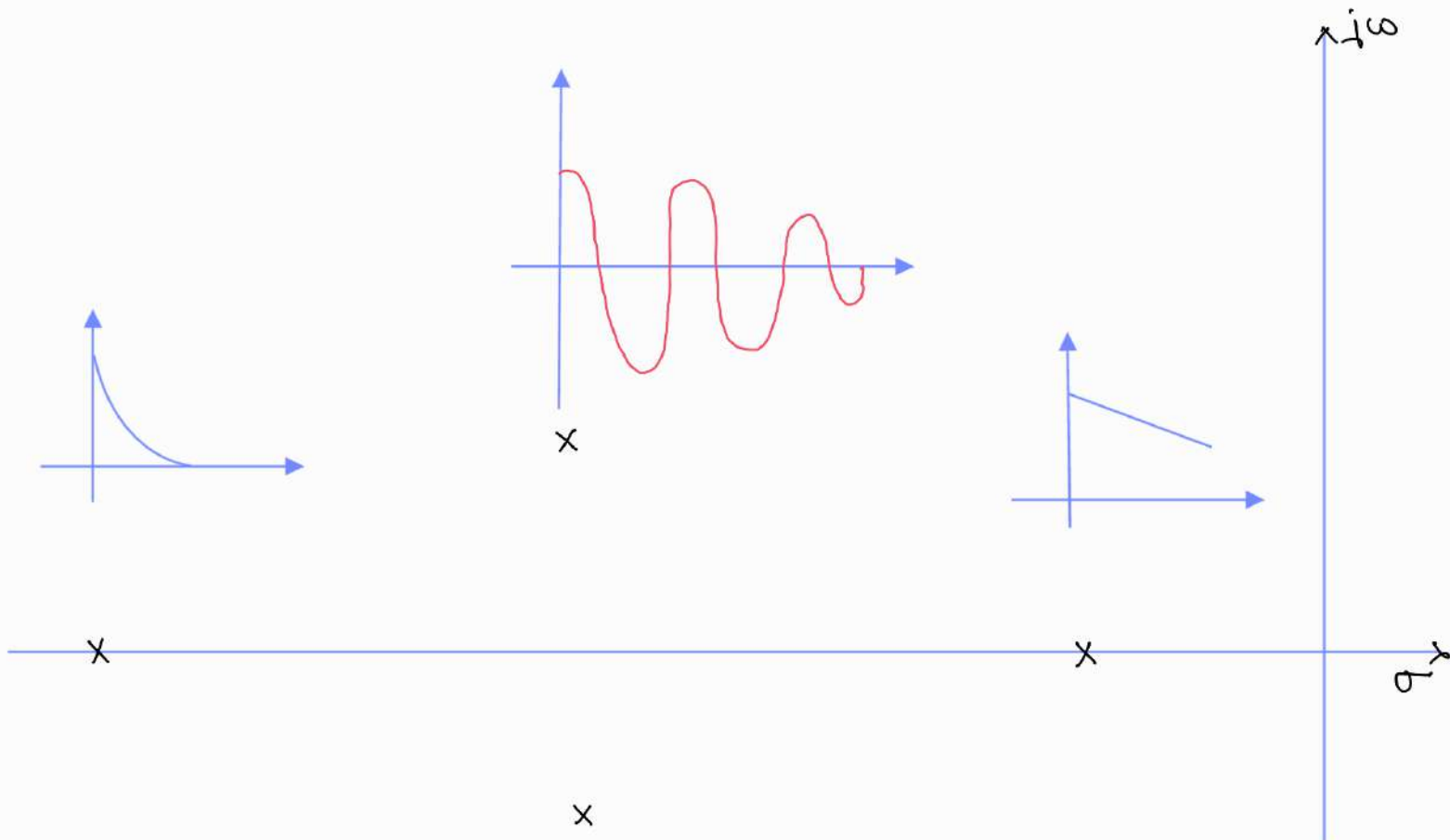
→ from  $s^1$  row: for the system to be stable  $\Rightarrow (6-k)/3 > 0$

→ for  $(6-k)/3 > 0 \Rightarrow k < 6$

→ range of  $K$  for the system to be stable is  $0 < k < 6$

## Relative stability

- indicates the closeness of the system to stable region.
- an indication of the strength or degree of stability.
- in time domain, the relative stability may be measured by relative settling times of each root or pair of roots.
- settling time is inversely proportional to the location of roots of characteristic equation.
- if the roots are located far away from imaginary axis  $\Rightarrow$  transient dies out faster  $\Rightarrow$  relative stability of the system will improve.



s-plane

$j\omega$

← more stable

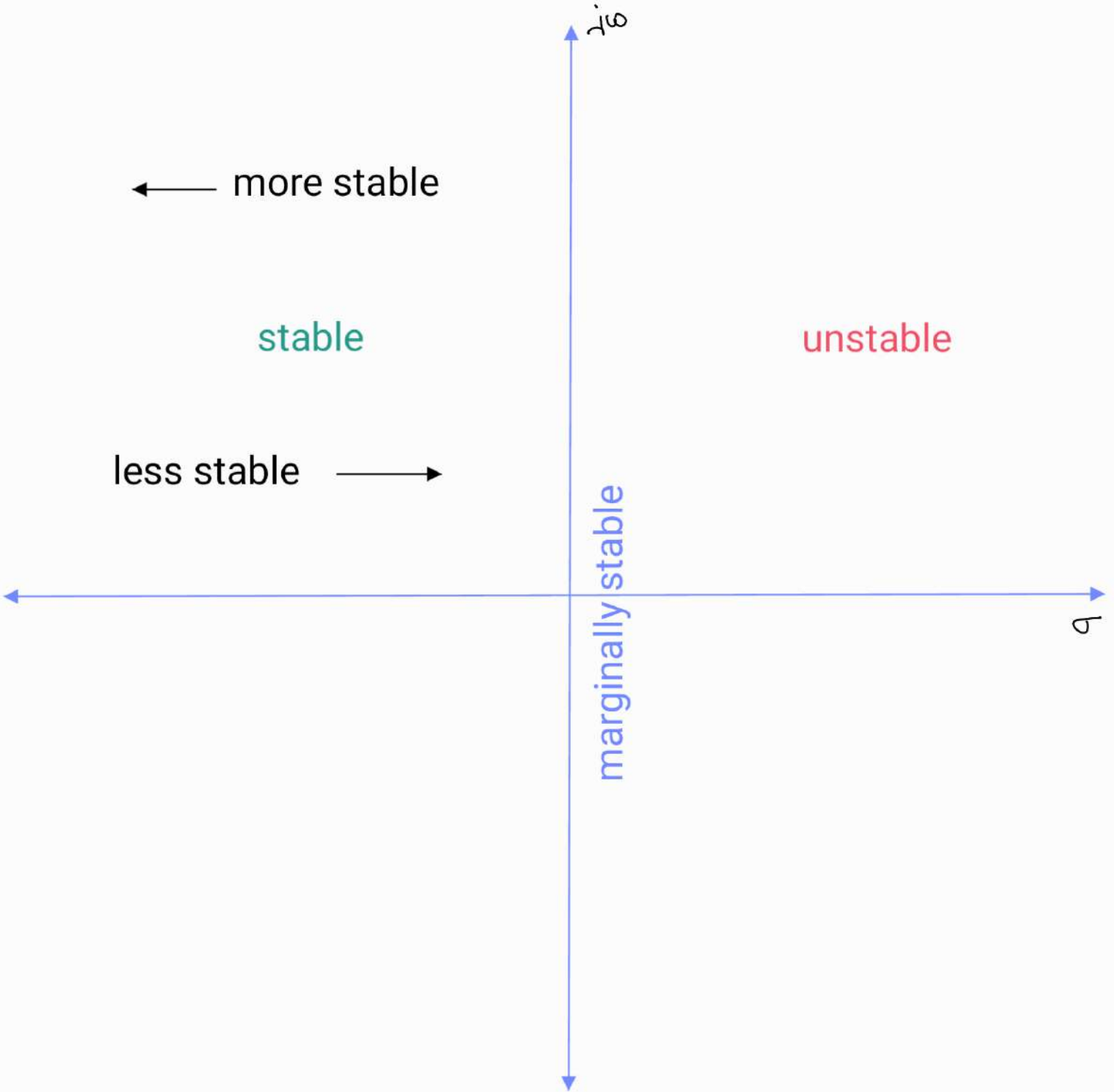
stable

unstable

less stable →

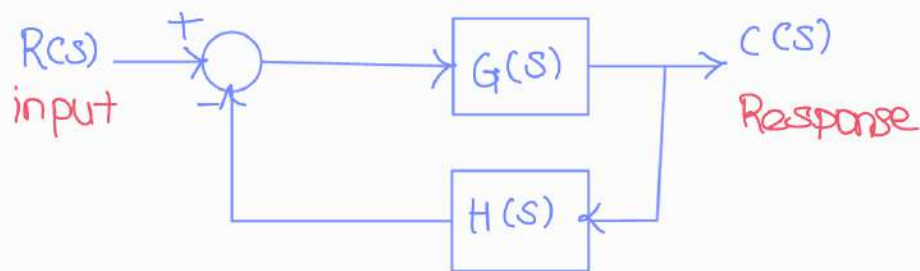
marginally stable

$\sigma$



# Time Response

- output of the closed loop system as a function of time.
- denoted by  $C(t)$ .
- obtained by solving the differential equation governing the system (or) from the transfer function and input of the system.
- in s-domain:



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} \Rightarrow C(s) = \left[ \frac{G(s)}{1+G(s)H(s)} \right] \cdot R(s)$$

- in time domain:

$$C(t) = \mathcal{L}^{-1}\{C(s)\}$$

- consist two parts:

- ↳ **Transient response:** when the input changes from one state to another.
- ↳ **Steady state response:** when 't' approaches infinity

$$C(t) = C_{tr}(t) + C_{ss}(t)$$

## Test Input

- Knowledge of input signal is required to predict the response of a system.
- In most of the systems, the input signals are not known ahead of time and also it is difficult to express the input signals mathematically by simple equations.
- The characteristics of actual input signals are a sudden change, a constant velocity, a constant acceleration and a sudden shock.
- Test signals which resembles these characteristics are used as input signals to predict the performance of the system.
- Standard test signals:
  - ↳ step
  - ↳ ramp
  - ↳ parabolic
  - ↳ impulse

→ Step signal:

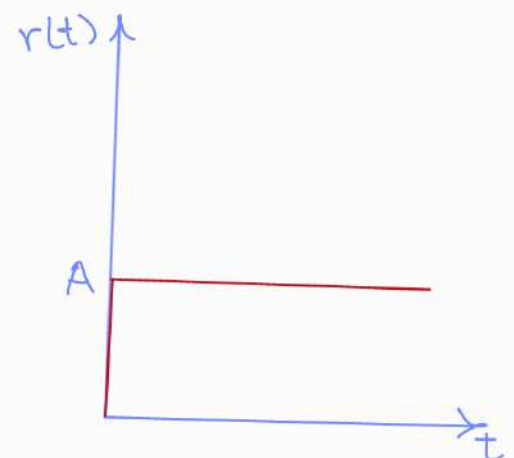
$$r(t) = A u(t)$$

$$u(t) = \begin{cases} 1 & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$

$$L[r(t)] = L[A u(t)] = \frac{A}{s}$$

unit step signal

$$A=1 \Rightarrow L[u(t)] = \frac{1}{s}$$

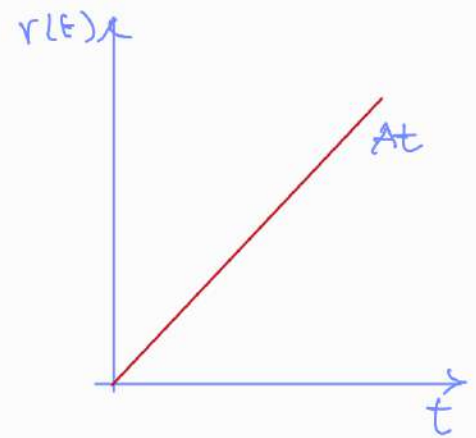


→ Ramp signal:

↳ Integral of step signal

$$r(t) = \begin{cases} At & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$

$$L[r(t)] = L[At] = A/s^2$$



**unit ramp signal**

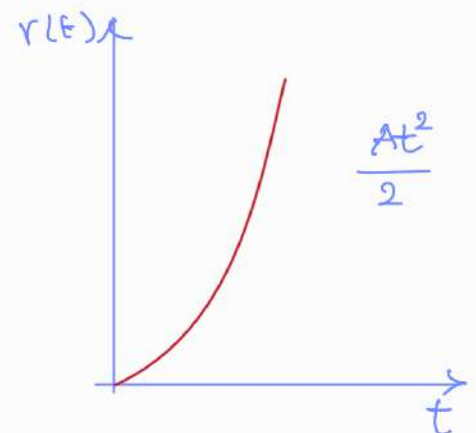
$$A=1 \Rightarrow L[r(t)] = 1/s^2$$

→ Parabolic signal:

↳ Integral of ramp signal.

$$r(t) = \begin{cases} At^2/2 & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$

$$L[r(t)] = L\left[\frac{At^2}{2}\right] = \frac{A}{s^3}$$



**unit parabolic signal**

$$A=1 \Rightarrow L[r(t)] = \frac{1}{s^3}$$

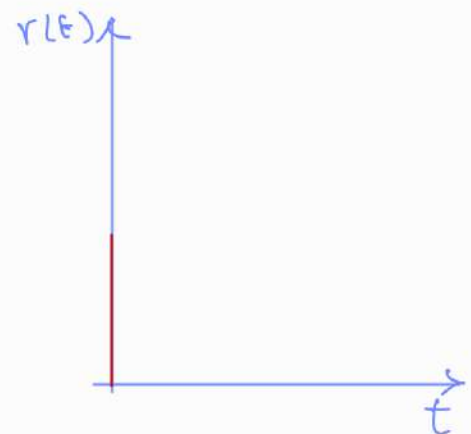
→ Impulse signal:

↳ derivative of step signal.

$$r(t) = \delta(t)$$

$$\delta(t) = \begin{cases} 1 & ; t = 0 \\ 0 & ; t \neq 0 \end{cases}$$

$$L[r(t)] = L[\delta(t)] = 1$$



## Order of a System

→ order of the differential equation governing the system.

$$\begin{aligned}
 & \text{①} \rightarrow n^{\text{th}} \text{ order system} \\
 & a_0 \frac{d^n p(t)}{dt^n} + a_1 \frac{d^{n-1} p(t)}{dt^{n-1}} + \dots + a_{n-1} \frac{d p(t)}{dt} + a_n p(t) = \\
 & \quad \text{Output/response} \\
 & b_0 \frac{d^m q(t)}{dt^m} + b_1 \frac{d^{m-1} q(t)}{dt^{m-1}} + \dots + b_{m-1} \frac{d q(t)}{dt} + b_m q(t) \\
 & \quad \text{Input/excitation}
 \end{aligned}$$

→ maximum power of  $s$  in the denominator polynomial of transfer function.

$$T(s) = \frac{P(s)}{Q(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_{m-1} s + b_m}{a_0 \text{①} s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

$n^{\text{th}} \text{ order system}$

→ number of poles in the transfer function.

$$T(s) = \frac{p(s)}{q(s)} = \frac{(s+z_1)(s+z_2)\dots(s+z_m)}{(s+p_1)(s+p_2)\dots(s+p_n) \text{①}}$$

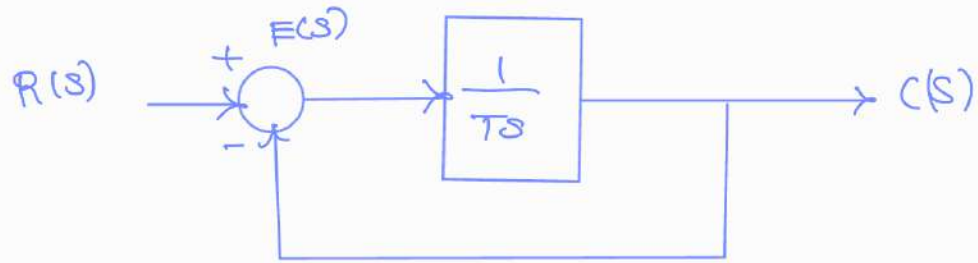
$n^{\text{th}} \text{ order system}$

$n = 0 \rightarrow$  The system is Zero order system

$n = 1 \rightarrow$  The system is First order system

$n = 2 \rightarrow$  The system is Second order system

## First order system



$$T(s) = \frac{C(s)}{R(s)}$$

$$= \frac{1/Ts}{1 + (1/Ts) \cdot 1}$$

$T \Rightarrow$  time Constant

$$= \frac{1/Ts}{\frac{Ts + 1}{Ts}}$$

$$= \frac{1}{Ts + 1}$$

$$= \frac{1}{T(s + \frac{1}{T})}$$

$$T(s) = \frac{1/T}{s + 1/T}$$

$$\frac{C(s)}{R(s)} = \frac{1/T}{s + 1/T}$$

$$C(s) = \left[ \frac{1/T}{(s + 1/T)} \right] R(s)$$

## Response of first order system for unit step signal

unit step input:  $r(t) = 1$

$$R(s) = \frac{1}{s}$$

$$C(s) = \left[ \frac{1/T}{s + 1/T} \right] R(s)$$

$$= \left[ \frac{1/T}{s + 1/T} \right] \cdot \frac{1}{s}$$

$$C(s) = \frac{1}{s} \cdot \frac{1/T}{s + 1/T}$$

partial fraction:

$$C(s) = \frac{A}{s} + \frac{B}{s + 1/T}$$

$$\frac{1}{s} \cdot \frac{1/T}{s + 1/T} = \frac{A}{s} + \frac{B}{s + 1/T}$$

$$\frac{1/T}{s(s + 1/T)} = \frac{A(s + 1/T) + Bs}{s(s + 1/T)}$$

$$\frac{1}{T} = A\left(s + \frac{1}{T}\right) + Bs$$

$$\text{put } s = -\frac{1}{T}$$

$$\frac{1}{T} = A(0) + B(-1/T)$$

$$B = -1$$

$$\text{put } s = 0$$

$$\frac{1}{T} = A\left(\frac{1}{T}\right) + B(0)$$

$$A = 1$$

$$C(s) = \frac{A}{s} + \frac{B}{s+1/\tau}$$

$$C(s) = \frac{1}{s} - \frac{1}{s+1/\tau}$$

Inverse Laplace transform:

$$\mathcal{L}^{-1}[C(s)] = \mathcal{L}^{-1}\left[\frac{1}{s} - \frac{1}{s+1/\tau}\right]$$

$$C(t) = 1 - e^{-t/\tau}$$

when

$$t=0 ; C(t) = 0$$

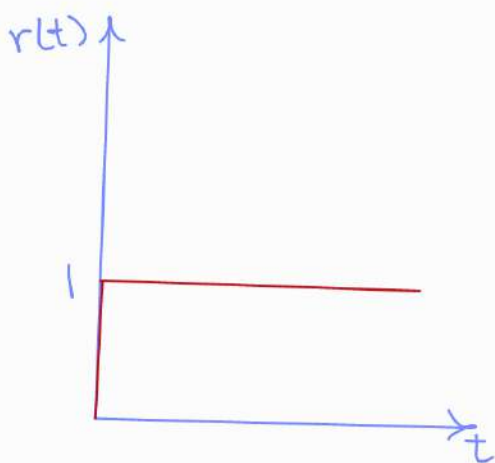
$$t=T ; C(t) = 0.632$$

$$t=2T ; C(t) = 0.865$$

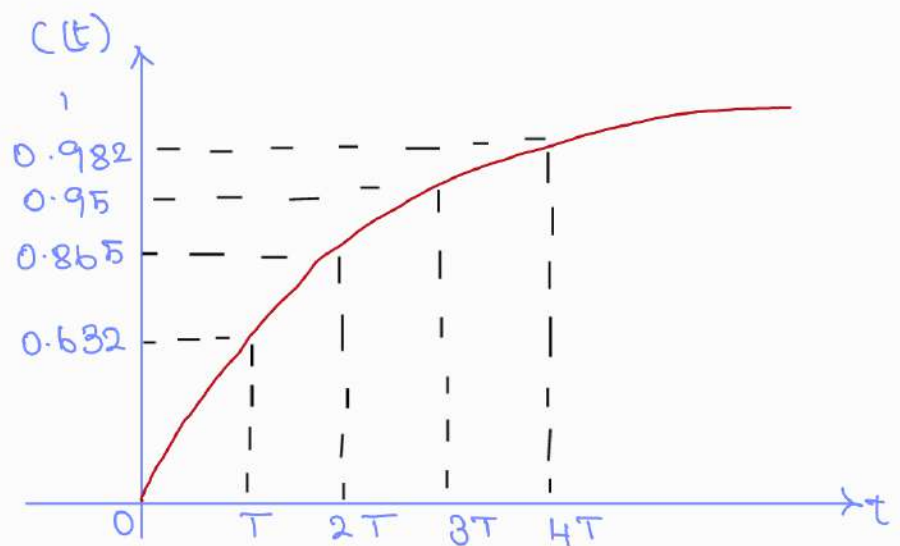
$$t=3T ; C(t) = 0.950$$

$$t=4T ; C(t) = 0.982$$

$$t=\infty ; C(t) = 1$$



unit step input



response for unit step input

## Response of first order system for unit impulse signal

unit impulse input:  $r(t) = \delta(t)$

$$R(s) = 1$$

$$C(s) = \frac{1/\tau}{s + 1/\tau} \cdot R(s)$$

$$= \frac{1/\tau}{s + 1/\tau} \cdot 1$$

$$C(s) = \frac{1/\tau}{s + 1/\tau}$$

Inverse Laplace Transform:

$$C(t) = \frac{1}{\tau} \cdot e^{-t/\tau}$$

when

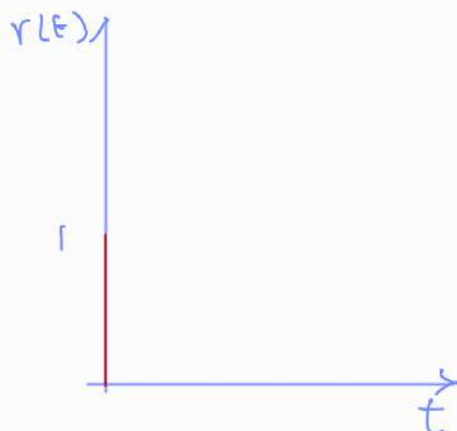
$$t = 0 ; C(t) = 1/\tau$$

$$t = \tau ; C(t) = 0.367/\tau$$

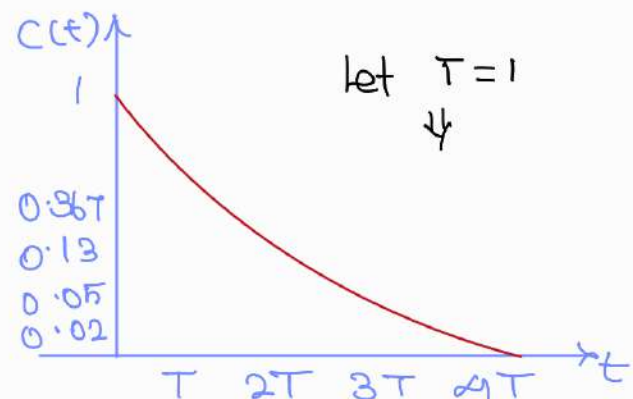
$$t = 2\tau ; C(t) = 0.13/\tau$$

$$t = 3\tau ; C(t) = 0.05/\tau$$

$$t = 4\tau ; C(t) = 0.02/\tau$$



unit impulse input



response for unit impulse input

## Response of first order system for unit ramp signal

unit ramp input:  $r(t) = t$

$$R(s) = 1/s^2$$

$$C(s) = \frac{1/\tau}{s + 1/\tau} \cdot R(s)$$

$$C(s) = \frac{1}{s^2} \cdot \frac{1/\tau}{s + 1/\tau}$$

partial fraction:

$$C(s) = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s + 1/\tau}$$

$$\frac{1/\tau}{s^2 \cdot (s + 1/\tau)} = \frac{A \cdot s \cdot (s + 1/\tau) + B(s + 1/\tau) + C s^2}{s^2 (s + 1/\tau)}$$

$$\frac{1}{\tau} = A s (s + \frac{1}{\tau}) + B(s + 1/\tau) + C s^2$$

put  $s=0$

$$\frac{1}{\tau} = A(0) + B(1/\tau) + C(0)$$

$$\frac{1}{\tau} = \frac{B}{\tau}$$

$$B = 1$$

put  $s = -1/\tau$

$$\frac{1}{\tau} = A(0) + B(0) + C(1/\tau^2)$$

$$\frac{1}{\tau} = \frac{C}{\tau^2}$$

$$C = \tau$$

put  $s = 1$

$$\frac{1}{\tau} = A\left(\tau + \frac{1}{\tau}\right) + \left(\tau + \frac{1}{\tau}\right) + \tau$$

$$\frac{1}{\tau} = \frac{A(\tau + 1) + (\tau + 1) + \tau^2}{\tau}$$

$$1 = A(\tau + 1) + (\tau + 1) + \tau^2$$

$$1 - \tau^2 = (A + 1)(\tau + 1)$$

$$A + 1 = (1 - \tau^2) / (\tau + 1)$$

$$A = (1 - \tau^2) / (\tau + 1) - 1$$

$$= 1 - \tau^2 - \tau - 1 / (\tau + 1)$$

$$= -\tau(1 + \tau) / (1 + \tau)$$

$$A = -\tau$$

$$C(s) = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1/T}$$

$$C(s) = -\frac{T}{s} + \frac{1}{s^2} + \frac{T}{s+1/T}$$

Inverse Laplace Transform:

$$C(t) = -T + t + T e^{-t/T}$$

$$C(t) = t - T + T e^{-t/T}$$

When

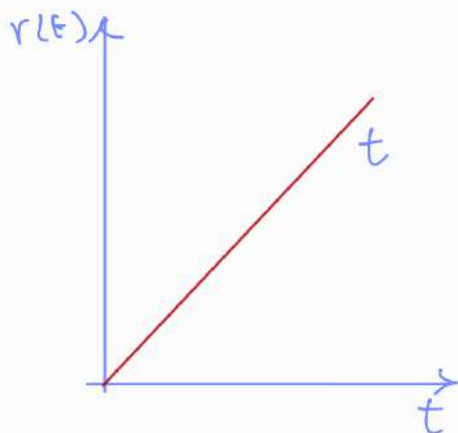
$$t=0 ; C(t) = 0$$

$$t=T ; C(t) = 0.367 T$$

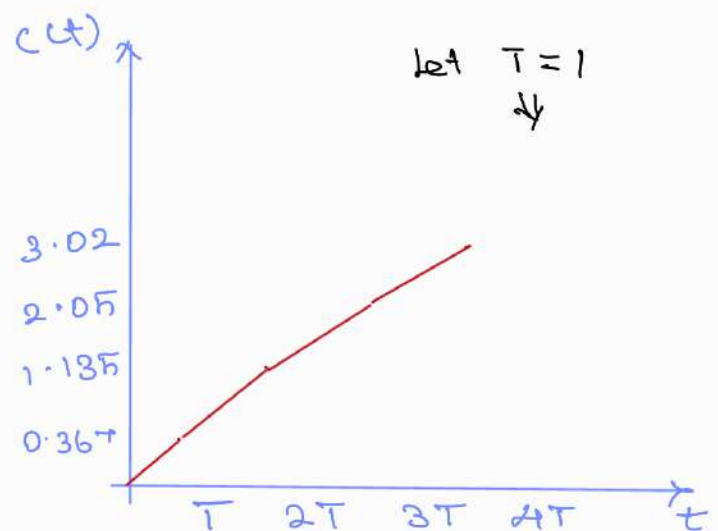
$$t=2T ; C(t) = 1.135 T$$

$$t=3T ; C(t) = 2.05 T$$

$$t=4T ; C(t) = 3.02 T$$



unit ramp input



response for unit ramp input

1. Find the open loop steady state gain and time constant for the following first order system.

$$i) \quad G(s) = \frac{4}{s+1}$$

$$\text{T.F of 1st os: } G(s) = \frac{K}{Ts+1}$$

$$\frac{K}{Ts+1} = \frac{4}{s+1}$$

$K \rightarrow$  steady state gain

$T \rightarrow$  time constant

$$K = 4$$

$$T = 1$$

$$ii) \quad G(s) = \frac{8}{4s+1}$$

$$\frac{K}{Ts+1} = \frac{8}{4s+1}$$

$$K = 8$$

$$T = 4$$

$$iii) \quad G(s) = \frac{1}{2(s+1/2)}$$

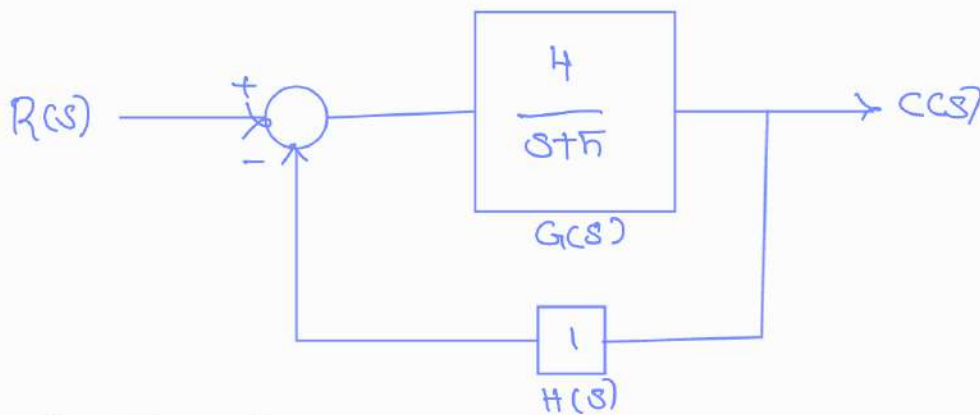
$$\frac{K}{Ts+1} = \frac{1}{2(s+1/2)}$$

$$\frac{K}{Ts+1} = \frac{1}{2s+1}$$

$$K = 1$$

$$T = 2$$

2. Obtain the response of unity feedback system when the input is (i) unit step and (ii) unit impulse.



Transfer Function:

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{G(s)}{1 + G(s)H(s)} \\ &= \frac{\frac{4}{s+5}}{1 + \left(\frac{4}{s+5}\right) \cdot 1} \\ &= \frac{4/(s+5)}{(s+5+4)/(s+5)}\end{aligned}$$

$$\frac{C(s)}{R(s)} = \frac{4}{s+9}$$

i) for unit step input:

$$R(s) = 1/s$$

$$C(s) = \left(\frac{4}{s+9}\right) \cdot R(s)$$

$$= \left(\frac{4}{s+9}\right) \cdot \frac{1}{s}$$

$$C(s) = \frac{1}{s} \cdot \frac{4}{s+9}$$

$$C(s) = \frac{A}{s} + \frac{B}{s+9}$$

$$4 = A(s+9) + Bs$$

$$\text{put } s=0$$

$$4 = A(9)$$

$$A = \frac{4}{9}$$

$$\text{put } s=-9$$

$$4 = B(-9)$$

$$B = \frac{-4}{9}$$

$$C(s) = \frac{4}{9s} - \frac{4}{9(s+9)}$$

Inverse Laplace transform:

$$C(t) = \frac{4}{9} - \frac{4}{9} e^{-9t}$$

$$C(t) = \frac{4}{9} (1 - e^{-9t})$$

ii) for a unit impulse input:

$$R(s) = 1$$

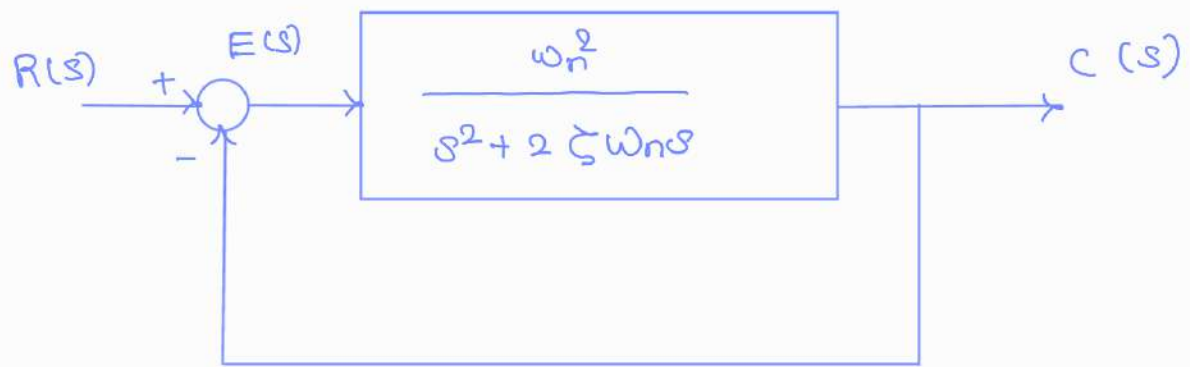
$$C(s) = \frac{4}{s+9} \cdot R(s)$$

$$C(s) = \frac{4}{s+9}$$

Inverse Laplace transform:

$$C(t) = 4 \cdot e^{-9t}$$

## Second order system



Transfer Function:

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s}}{1 + \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s}} \\ &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s} \cdot \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2}\end{aligned}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot R(s)$$

$\omega_n \rightarrow$  undamped natural frequency

$\zeta \rightarrow$  damping ratio { actual damping / critical damping

$\hookrightarrow \zeta = 0 \Rightarrow$  undamped system.

$0 < \zeta < 1 \Rightarrow$  under damped system

$\zeta = 1 \Rightarrow$  critically damped system

$\zeta > 1 \Rightarrow$  over damped system

3. Find the impulse response for the given second order system.

$$\frac{C(s)}{R(s)} = \frac{5}{s^2 + 8s + 16}$$

impulse signal:

$$R(s) = 1$$

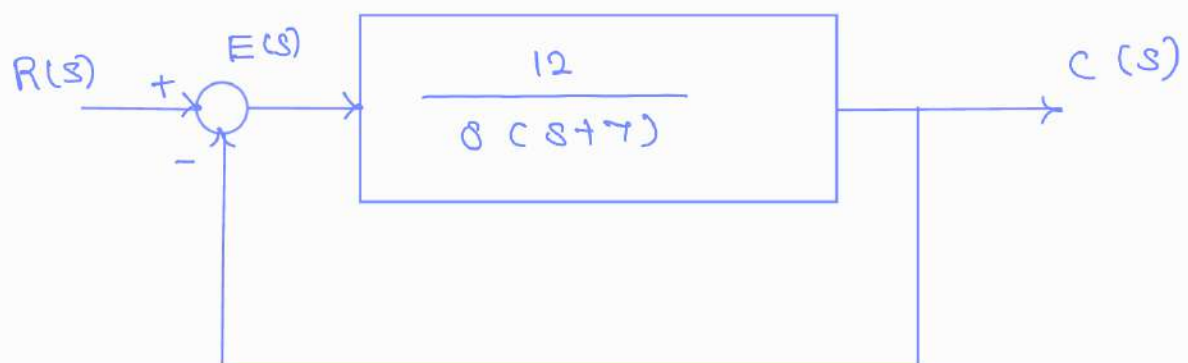
$$C(s) = \frac{5}{s^2 + 8s + 16}$$

$$C(s) = \frac{5}{(s+4)^2}$$

Inverse Laplace transform:

$$C(t) = 5 \cdot t \cdot e^{-4t}$$

4. Obtain the response of the system whose block diagram is given below.



Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{12/s(s+7)}{1 + \left( \frac{12}{s(s+7)} \cdot 1 \right)}$$

$$= \frac{12}{s^2 + 7s + 12}$$

$$\frac{C(s)}{R(s)} = \frac{12}{(s+4)(s+3)}$$

$$C(s) = \frac{12}{(s+4)(s+3)} \cdot R(s)$$

i) for unit step signal:

$$R(s) = 1/s$$

$$C(s) = \frac{1}{s} \cdot \frac{12}{(s+4)(s+3)}$$

$$C(s) = \frac{A}{s} + \frac{B}{(s+4)} + \frac{C}{(s+3)}$$

$$12 = A(s+4)(s+3) + B \cdot s(s+3) + C(s+4)$$

|                                                                  |                                                                                |                                                                                |
|------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <p>put <math>s=0</math></p> $12 = A(4)(3) + B(0) + C(0)$ $A = 1$ | <p>put <math>s=-4</math></p> $12 = A(0) + B(4)(-1) + C(0)$ $12 = -4B$ $B = -3$ | <p>put <math>s=-3</math></p> $12 = A(0) + B(0) + C(-3)(1)$ $12 = -3C$ $C = -4$ |
|------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|

$$C(s) = \frac{1}{s} + \frac{3}{(s+4)} - \frac{4}{(s+3)}$$

Inverse Laplace transform:

$$c(t) = 1 + 3e^{-4t} - 4e^{-3t}$$

ii) for unit impulse signal:

$$R(s) = 1$$

$$C(s) = \frac{12}{(s+4)(s+3)}$$

$$C(s) = \frac{A}{(s+4)} + \frac{B}{(s+3)}$$

$$12 = A(s+3) + B(s+4)$$

$$\text{put } s = -3$$

$$12 = A(0) + B(1)$$

$$B = 12$$

$$\text{put } s = -4$$

$$12 = A(-1) + B(0)$$

$$A = -12$$

$$C(s) = \frac{-12}{(s+4)} + \frac{12}{(s+3)}$$

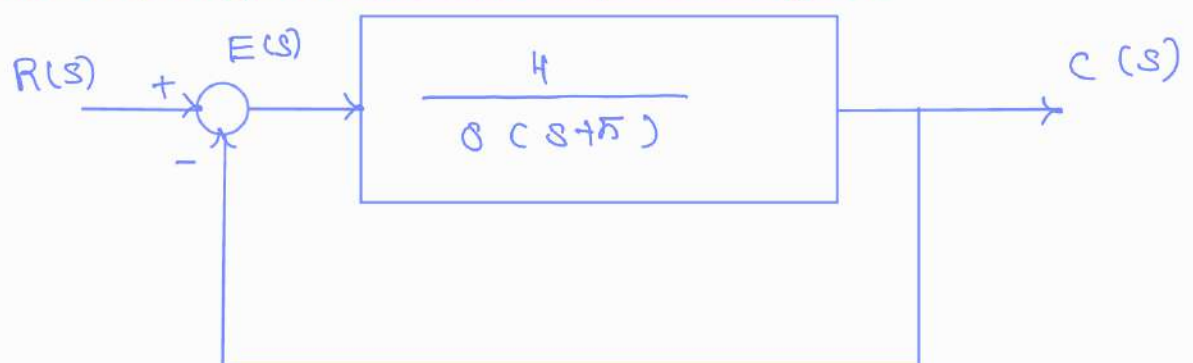
$$C(s) = \frac{12}{s+3} - \frac{12}{s+4}$$

Inverse Laplace transform:

$$C(t) = 12 e^{-3t} - 12 e^{-4t}$$

$$C(t) = 12(e^{-3t} - e^{-4t})$$

5. Obtain the response of unity feedback system when the input is unit step signal and unit impulse signal.



Transfer function:

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{G(s)}{1 + G(s)H(s)} \\ &= \frac{4/s(s+5)}{1 + \frac{4}{s(s+5)} \cdot (1)} \\ &= \frac{4/s(s+5)}{(s(s+5) + 4)/s(s+5)} \\ &= \frac{4}{s^2 + 5s + 4}\end{aligned}$$

$$\frac{C(s)}{R(s)} = \frac{4}{(s+1)(s+4)}$$

$$C(s) = \frac{4}{(s+1)(s+4)} \cdot R(s)$$

i) unit step input:

$$R(s) = 1/s$$

$$C(s) = \frac{1}{s} \cdot \frac{4}{(s+1)(s+4)}$$

partial fraction:

$$\begin{aligned}C(s) &= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4} \\ \frac{4}{s(s+1)(s+4)} &= \frac{A(s+1)(s+4) + B(s)(s+4) + C(s)(s+1)}{s(s+1)(s+4)}\end{aligned}$$

$$4 = A(s+1)(s+4) + Bs(s+4) + Cs(s+1)$$

|                                                                   |                                                                         |                                                                         |
|-------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| put $s = 0$<br>$4 = A(1)(4) + B(0) + C(0)$<br>$4 = 4A$<br>$A = 1$ | put $s = -1$<br>$4 = A(0) + B(-1)(3) + C(0)$<br>$4 = -3B$<br>$B = -4/3$ | put $s = -4$<br>$4 = A(0) + B(0) + C(-4)(-3)$<br>$4 = 12C$<br>$C = 1/3$ |
|-------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|

$$C(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4}$$

$$C(s) = \frac{1}{s} + \frac{-4}{3} \left( \frac{1}{s+1} \right) + \frac{1}{3} \left( \frac{1}{s+4} \right)$$

Inverse Laplace Transform:

$$C(t) = 1 - \frac{4}{3} e^{-t} + \frac{1}{3} e^{-4t}$$

ii) unit impulse input:

$$R(s) = 1$$

$$C(s) = \frac{4}{(s+1)(s+4)} = \frac{A}{(s+1)} + \frac{B}{(s+4)}$$

partial fraction:

$$\frac{4}{(s+1)(s+4)} = \frac{A(s+4) + B(s+1)}{(s+1)(s+4)}$$

$$4 = A(s+4) + B(s+1)$$

|                                                |                                                  |
|------------------------------------------------|--------------------------------------------------|
| put $s = -1$<br>$4 = A(3) + B(0)$<br>$A = 4/3$ | put $s = -4$<br>$4 = A(0) + B(-3)$<br>$B = -4/3$ |
|------------------------------------------------|--------------------------------------------------|

$$C(s) = \frac{4}{3} \left( \frac{1}{s+1} \right) - \frac{4}{3} \left( \frac{1}{s+4} \right)$$

Inverse Laplace Transform:

$$C(t) = \frac{4}{3} [e^{-t} - e^{-4t}]$$

## Response of undamped Second order system

Transfer function:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

for undamped system:

$$\zeta = 0$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + \omega_n^2}$$

for unit step input:

$$R(s) = 1/s$$

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{s^2 + \omega_n^2}$$

partial fraction:

$$C(s) = \frac{A}{s} + \frac{B}{s^2 + \omega_n^2}$$

$$\frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{A(s^2 + \omega_n^2) + Bs}{s(s^2 + \omega_n^2)}$$

$$A(s^2 + \omega_n^2) + Bs = \omega_n^2$$

$$\text{put } s = 0$$

$$A(\omega_n^2) + B(0) = \omega_n^2$$

$$A = 1$$

$$\text{put } s^2 = -\omega_n^2 \Rightarrow s = j\omega_n$$

$$A(0) + B(j\omega_n) = \omega_n^2$$

$$B = -j\omega_n = -s$$

$$C(s) = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}$$

$$L[1] = 1/s$$

$$L[\cos at] = \frac{s}{s^2 + a^2}$$

Inverse Laplace transform:

$$C(t) = 1 - \cos \omega_n t$$

for unit impulse input:

$$R(s) = 1$$

$$C(s) = \frac{\omega_n^2}{s^2 + \omega_n^2} \cdot R(s)$$

$$C(s) = \frac{\omega_n^2}{s^2 + \omega_n^2} \cdot (1)$$

$$C(s) = \frac{\omega_n^2}{s^2 + \omega_n^2}$$

$$C(s) = \omega_n \left[ \frac{\omega_n}{s^2 + \omega_n^2} \right] \quad \left| \quad \mathcal{L}[\sin at] = \frac{a}{s^2 + a^2} \right.$$

Inverse Laplace transform:

$$C(t) = \omega_n \cdot \sin \omega_n t$$

## Response of critically damped Second order system

Transfer function:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

for critical damping:  $\zeta = 1$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s + \omega_n)^2}$$

for unit step input:

$$R(s) = 1/s$$

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s + \omega_n)^2}$$

partial fraction:

$$C(s) = \frac{A}{s} + \frac{B}{(s + \omega_n)} + \frac{C}{(s + \omega_n)^2}$$

$$\begin{aligned} A &= s \times C(s) \Big|_{s=0} \\ &= s \times \frac{\omega_n^2}{s(s + \omega_n)^2} \Big|_{s=0} \\ &= \frac{\omega_n^2}{\omega_n^2} \end{aligned}$$

$$A = 1$$

$$\begin{aligned} B &= \frac{d}{ds} \left[ (s + \omega_n)^2 \times C(s) \right] \Big|_{s = -\omega_n} \\ &= \frac{d}{ds} \left[ (s + \omega_n)^2 \times \frac{\omega_n^2}{s(s + \omega_n)^2} \right] \Big|_{s = -\omega_n} \\ &= \frac{d}{ds} \left[ \frac{\omega_n^2}{s} \right] \Big|_{s = -\omega_n} \\ &= -\frac{\omega_n^2}{s^2} \Big|_{s = -\omega_n} \end{aligned}$$

$$B = -1$$

$$\begin{aligned} C &= (s + \omega_n)^2 \times C(s) \Big|_{s = -\omega_n} \\ &= (s + \omega_n)^2 \times \frac{\omega_n^2}{s(s + \omega_n)^2} \\ &= \frac{\omega_n^2}{s} \Big|_{s = -\omega_n} \\ &= \frac{\omega_n^2}{-\omega_n} \\ C &= -\omega_n \end{aligned}$$

$$C(s) = \frac{1}{s} - \frac{1}{(s+\omega_n)} - \frac{\omega_n}{(s+\omega_n)^2}$$

$$\begin{aligned} \mathcal{L}[1] &= 1/s \\ \mathcal{L}[e^{-at}] &= \frac{1}{s+a} \\ \mathcal{L}[te^{-at}] &= \frac{1}{(s+a)^2} \end{aligned}$$

Inverse Laplace transform:

$$c(t) = 1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}$$

$$c(t) = [1 - e^{-\omega_n t} (1 + \omega_n t)]$$

for impulse input:

$$R(s) = 1$$

$$C(s) = \frac{\omega_n^2}{(s+\omega_n)^2} \cdot R(s)$$

$$C(s) = \frac{\omega_n^2}{(s+\omega_n)^2}$$

$$C(s) = \omega_n^2 \cdot \left( \frac{1}{(s+\omega_n)^2} \right)$$

$$\mathcal{L}[te^{-at}] = \frac{1}{(s+a)^2}$$

Inverse Laplace Transform:

$$c(t) = \omega_n^2 t e^{-\omega_n t}$$

for under damped second order system :

$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot R(s)$$

$$0 < \zeta < 1$$

for unit step input:

$$R(s) = 1/s$$

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1-\zeta^2})^2} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1-\zeta^2})^2}$$

Inverse Laplace transform:

$$C(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta)$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

$$\theta = \frac{\tan^{-1} \sqrt{1-\zeta^2}}{\zeta}$$

for over damped second order system:

$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot R(s)$$

$$\zeta > 1$$

for unit step input:

$$R(s) = 1/s$$

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s + s_1)(s + s_2)}$$

$$s_1 = \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$$

$$s_2 = \zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$$

$$C(s) = \frac{A}{s} + \frac{B}{(s + s_1)} + \frac{C}{(s + s_2)}$$

$$C(s) = \frac{1}{s} + \frac{1}{s_1} \frac{-\omega_n/2\sqrt{\zeta^2 - 1}}{(s + s_1)} + \frac{1}{s_2} \frac{\omega_n/2\sqrt{\zeta^2 - 1}}{(s + s_2)}$$

$$C(t) = 1 - \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \frac{1}{s_1} \left[ \frac{e^{-s_1 t}}{s_1} - \frac{e^{-s_2 t}}{s_2} \right]$$

⇒ roots are purely imaginary and the system is undamped:

$$\zeta = 0 \Rightarrow s_1, s_2 = \pm j\omega_n$$

⇒ roots are real and equal and the system is critically damped:

$$\zeta = 1 \Rightarrow s_1, s_2 = -\omega_n$$

⇒ roots are real and unequal and The system is over damped:

$$\zeta > 1 \Rightarrow s_1, s_2 = -\zeta \omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

⇒ roots are complex conjugate and the system is under damped:

$$0 < \zeta < 1 \Rightarrow s_1, s_2 = -\zeta \omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$$

6. The response of servo mechanism is given below, subject to a unit step input. Obtain an expression for closed loop transfer function and determine the undamped natural frequency and damping ratio.

$$C(t) = 1 + 0.2 e^{-60t} - 1.2 e^{-10t}$$

Laplace Transform:

$$C(s) = \frac{1}{s} + \frac{0.2}{(s+60)} - \frac{1.2}{(s+10)}$$

$$= \frac{(s+60)(s+10) + 0.2s(s+10) - 1.2s(s+60)}{s(s+60)(s+10)}$$

$$= \frac{s^2 + 70s + 600 + 0.2s^2 + 2s - 1.2s^2 - 72s}{s(s+60)(s+10)}$$

$$= \frac{1.2s^2 - 1.2s^2 + 72s - 72s + 600}{s(s^2 + 70s + 600)}$$

$$C(s) = \frac{600}{s(s^2 + 70s + 600)}$$

$$C(s) = \frac{1}{s} \cdot \frac{600}{s^2 + 70s + 600}$$

$$C(s) = R(s) \cdot \frac{600}{s^2 + 70s + 600}$$

$$\frac{C(s)}{R(s)} = \frac{600}{s^2 + 70s + 600} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n^2 = 600$$

$$\omega_n = \sqrt{600} = 24.49 \text{ rad/sec}$$

$$2\zeta\omega_n = 70$$

$$\zeta = \frac{70}{2} \times \frac{1}{24.49}$$

$$\zeta = 1.43$$

Natural frequency of oscillation = 24.49 rad/sec

Damping ratio = 1.43

## Root Locus Technique

→ graphical method in which the movement of roots in the s-plane is sketch when the gain of the system (K) is varied from 0 to infinity.

→ Consider unity feedback system  $G(s)$ . Obtain the root locus.

$$G(s) = \frac{K}{s}$$

characteristic equation:

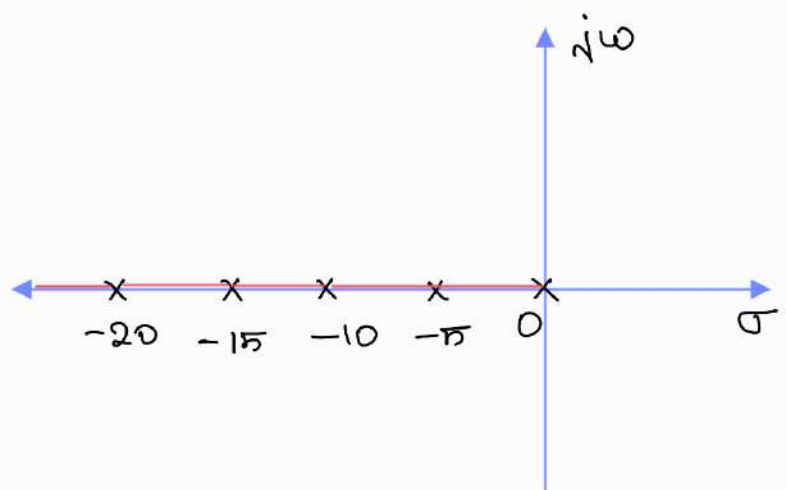
$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s} * 1 = 0$$

$$\frac{s+K}{s} = 0$$

$$s = -K$$

| K  | s   |
|----|-----|
| 0  | 0   |
| 5  | -5  |
| 10 | -10 |
| 15 | -15 |
| ⋮  | ⋮   |



→ tool for adjusting the location of closed loop poles to achieve The desired system performance by varying one or more system parameters.

→ the path taken by roots of characteristic equation when open loop gain  $K$  is varied from 0 to  $\infty \Rightarrow$  root locus/ loci

→ consider open loop transfer function:

$$G(s) = \frac{K}{s(s+p_1)(s+p_2)}$$

characteristic equation:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s+p_1)(s+p_2)} = 0$$

$$s(s+p_1)(s+p_2) + K = 0$$

$$\Rightarrow G(s)H(s) = -1$$

magnitude condition:

$$|G(s)H(s)| = 1$$

angle condition:

$$\angle G(s)H(s) = \pm 180^\circ (2q+1)$$

$q = 0, 1, 2, \dots$

→ roots of characteristic equation depend on open loop gain  $K$ .

→ when  $K$  is varied from 0 to  $\infty \Rightarrow$  roots of characteristic equation will take different values.

→ when  $K = 0 \Rightarrow$  the roots of characteristic equation are given by open loop poles.

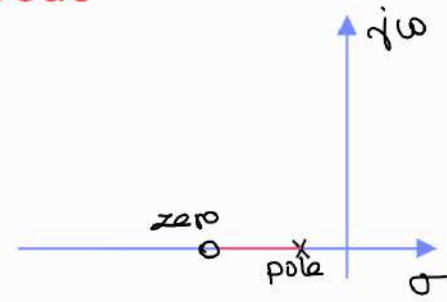
→ when  $K = \infty \Rightarrow$  the roots of characteristic equation are given by open loop zeros.

→ It is very difficult to plot the root locus to higher order system by substitution of  $K$  in the characteristic equation from 0 to infinity.

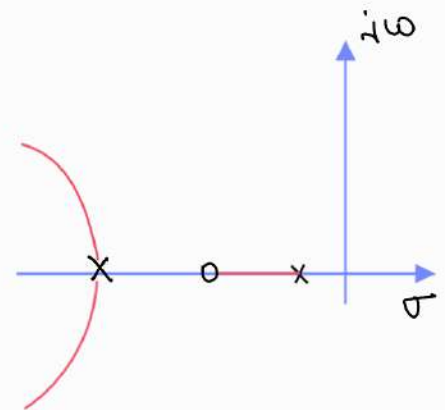
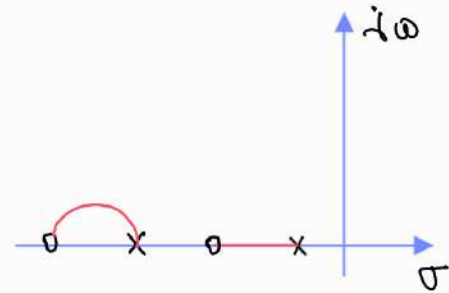
→ simplify the construction of root locus for the higher order system  $\Rightarrow$  root locus technique.

## Rules for construction of Root Locus

↳ the root locus is symmetrical about the real axis.



↳ the root locus originates from an open loop pole (when  $k=0$ ) and terminates at either on finite open loop zero or open loop zero at infinity.



↳ the number of branches of the root locus  $[N]$  while  $n \rightarrow$  number of poles and  $m \rightarrow$  number of zeros

$$\begin{aligned} \text{if } n > m &\Rightarrow N = n \\ m > n &\Rightarrow N = m \\ n = m &\Rightarrow N = n = m \end{aligned}$$

↳ the number of branches of the root locus terminating on infinity is equal to  $n - m$  [number of poles - number of zeros].

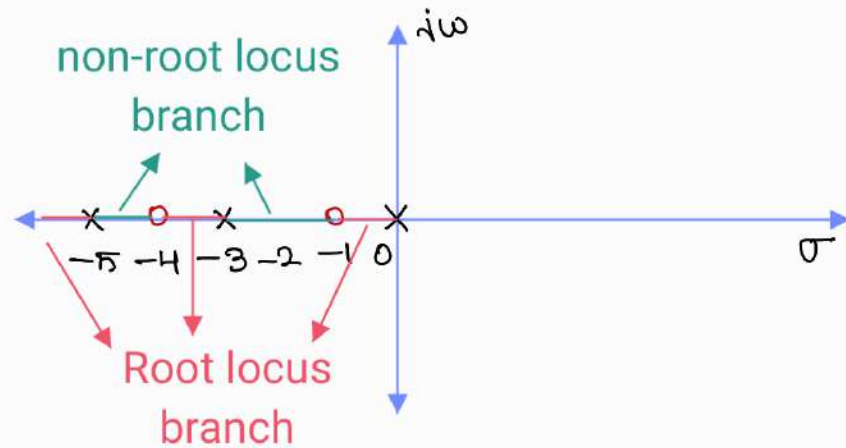
$$\frac{K}{s} \Rightarrow n - m = 1 - 0 = 1$$

$$\frac{K}{s(s+1)} \Rightarrow n - m = 2 - 0 = 2$$

$$\frac{s}{s(s+1)} \Rightarrow n - m = 2 - 1 = 1$$

↳ sum of the number of open loop poles and open loop zeros on the real axis to the right hand side of a point is odd number  $\Rightarrow$  that point on real axis lies on the root locus.

$$\frac{K(s+1)(s+4)}{s(s+3)(s+5)}$$



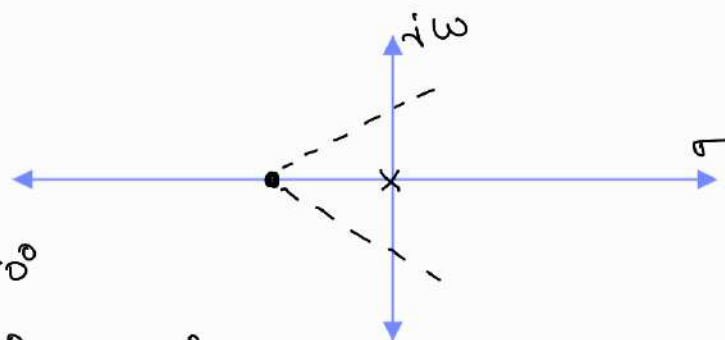
↳ angle of asymptotes with real axis:

$$\text{If } n > m \Rightarrow \theta_A = \frac{(2q+1)180^\circ}{n-m}; q = 0, 1, 2, \dots, n-m-1$$

$$m > n \Rightarrow \theta_A = \frac{(2q+1)180^\circ}{m-n}; q = 0, 1, 2, \dots, m-n-1$$

$(n - m)$  branches will approach the infinity along straight line  $\Rightarrow$  asymptotes.

| $n-m$ | Asymptotes | Angle                                                                                   |
|-------|------------|-----------------------------------------------------------------------------------------|
| 0     | 0          | -                                                                                       |
| 1     | 1          | $\theta = 180^\circ$                                                                    |
| 2     | 2          | $\theta_1 = 90^\circ; \theta_2 = 270^\circ$                                             |
| 3     | 3          | $\theta_1 = 60^\circ; \theta_2 = 180^\circ; \theta_3 = 300^\circ$                       |
| 4     | 4          | $\theta_1 = 45^\circ; \theta_2 = 135^\circ; \theta_3 = 225^\circ; \theta_4 = 315^\circ$ |



↳ centroid  $\Rightarrow$  point of intersection of the asymptotes with real axis.

$$\sigma_A = \frac{\sum \text{real part of pole} - \sum \text{real part of zero}}{n - m}$$

$$\frac{k}{(s+1)(s+2+2j)(s+2-2j)}$$

$$n = 3 ; m = 0 \Rightarrow N = 3$$

$$\theta_A = \frac{(2q+1)180^\circ}{3} ; q = 0, 1, 2$$

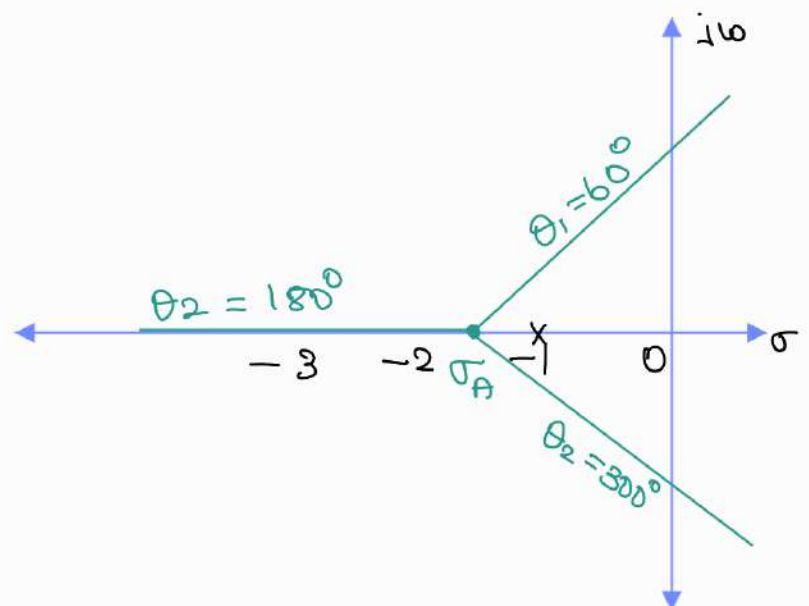
$$\theta_1 = 60^\circ ; \theta_2 = 180^\circ ; \theta_3 = 300^\circ$$

$$s = -1, -2-2j, -2+2j$$

$$\sigma_A = \frac{[-1 + (-2) + (-2)] - [0]}{3 - 0}$$

$$\sigma_A = -\frac{5}{3}$$

$$\sigma_A = -1.67$$



↳ the break away and breakin points of the root locus are determined from the roots of the equation  $(dK / ds) = 0$ .

breakaway angle  $\theta_b = \pm \frac{180^\circ}{r}$  number of branches approaching / leaving

$$G(s) H(s) = \frac{K}{s(s+1)(s+4)}$$

$$1 + G(s) H(s) = 0$$

$$1 + \frac{K}{s(s+1)(s+4)} = 0$$

$$\frac{s(s+1)(s+4) + K}{s(s+1)(s+4)} = 0$$

$$(s^2 + s)(s+4) + K = 0$$

$$s^3 + 5s^2 + 4s + K = 0$$

$$K = -s^3 - 5s^2 - 4s$$

$$\frac{dK}{ds} = 0$$

$$-3s^2 - 10s - 4 = 0$$

$$3s^2 + 10s + 4 = 0$$

$$s = -0.46, -2.86$$

$$K = -(s^3 + 5s^2 + 4s)$$

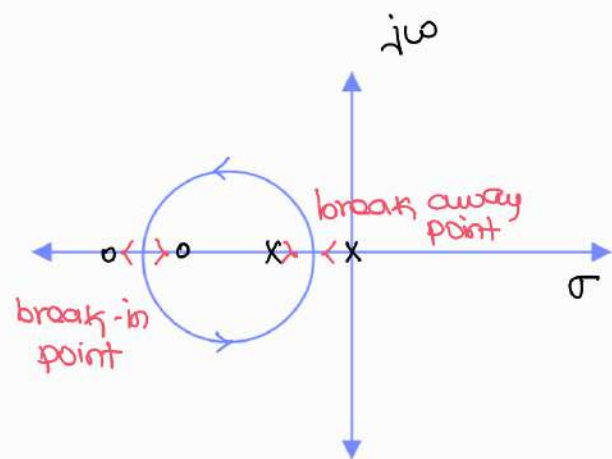
$$s = -0.46$$

$$K = 0.879$$

$$s = -2.86$$

$$K = -6.064$$

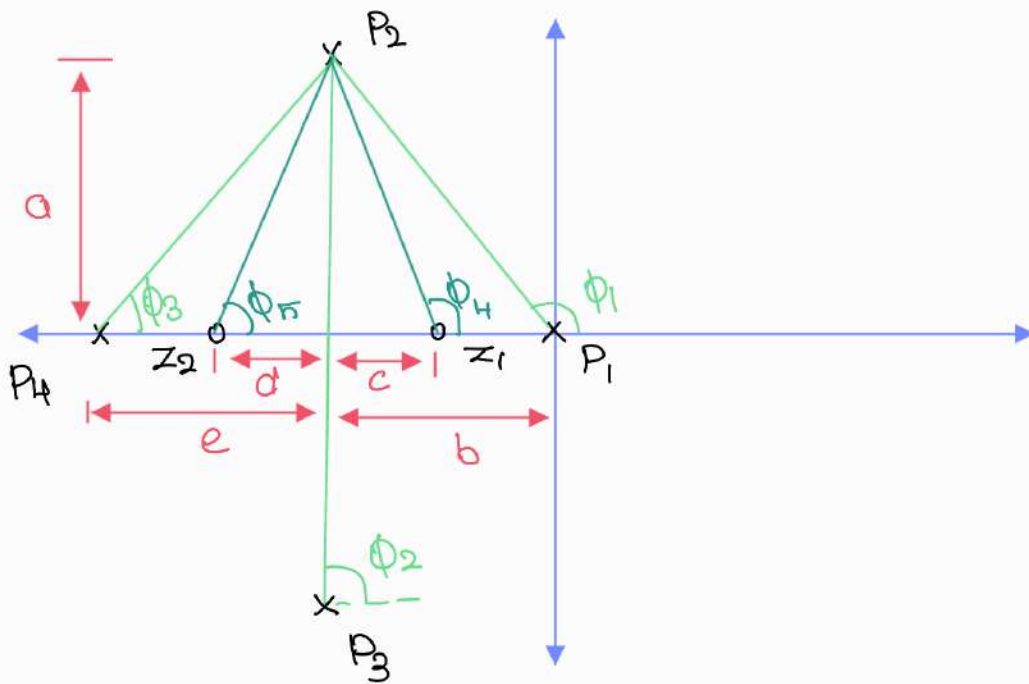
$$\theta_b = \frac{180^\circ}{2} = 90^\circ$$



↳ angle of departure from a complex open loop pole:

$$\phi_d = \pm 180^\circ + \phi$$

$\phi \rightarrow$  net angle contribution at the pole by all other open loop poles and zeros.



$$\phi_1 = 180^\circ - \tan^{-1} \frac{a}{b}$$

$$\phi_2 = 90^\circ$$

$$\phi_3 = \tan^{-1} \frac{a}{e}$$

$$\phi_4 = 180^\circ - \tan^{-1} \frac{a}{c}$$

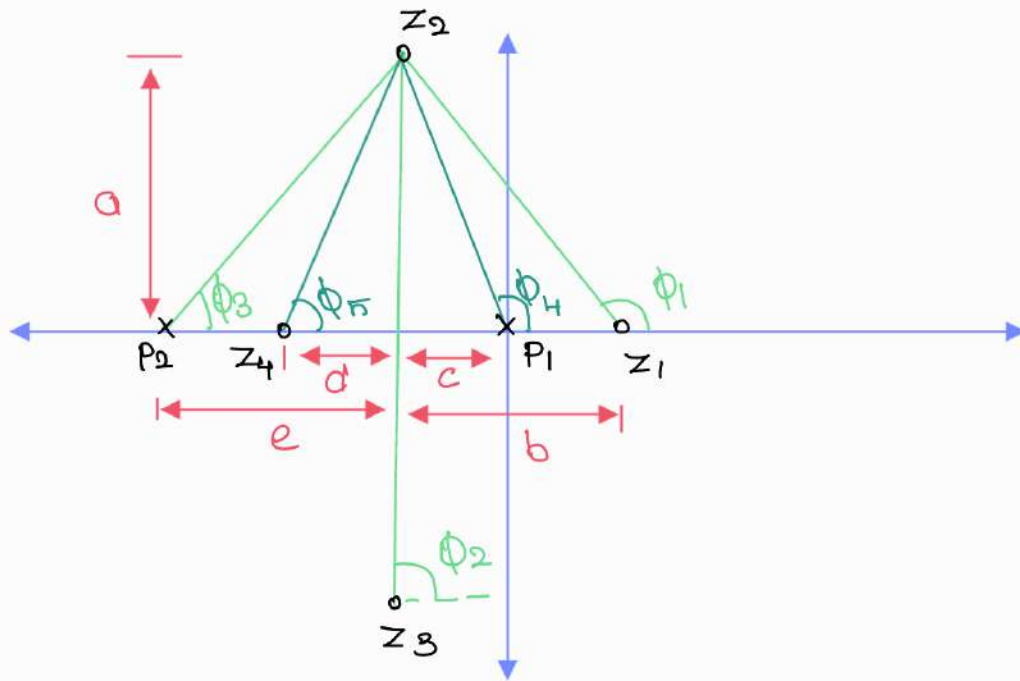
$$\phi_5 = \tan^{-1} \frac{a}{d}$$

$$\text{at } P_2 \Rightarrow \phi_d = 180^\circ - [(\phi_1 + \phi_2 + \phi_3) + (\phi_4 + \phi_5)]$$

↳ angle of arrival at a complex open loop zero:

$$\phi_o = \pm 180^\circ + \phi$$

$\phi \rightarrow$  net angle contribution at the zero by all other open loop poles and zeros.



$$\phi_1 = 180^\circ - \tan^{-1} \frac{a}{b}$$

$$\phi_2 = 90^\circ$$

$$\phi_3 = \tan^{-1} \frac{a}{e}$$

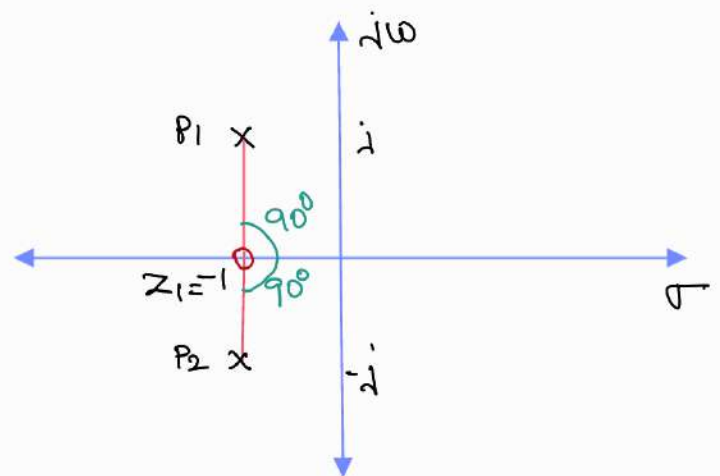
$$\phi_4 = 180^\circ - \tan^{-1} \frac{a}{c}$$

$$\phi_5 = \tan^{-1} \frac{a}{d}$$

$$\text{at } Z_2 \Rightarrow \phi_a = 180^\circ - [(\phi_1 + \phi_2 + \phi_3) + (\phi_4 + \phi_5)]$$

$$\frac{K(s+1)}{s^2+2s+2}$$

$$\begin{array}{l|l} s \rightarrow -1+j, -1-j & s = -1 \\ P_1 = -1+j & Z_1 = -1 \\ P_2 = -1-j & \end{array}$$



$$\phi = 90^\circ - 90^\circ = 0$$

$$\phi_a = 180^\circ + 0 = 180^\circ$$

↳ the points of intersection of root locus branches with the imaginary axis can be determined by Routh criterion.

- construct the characteristic equation
- construct the Routh array in terms of K
- determine the K margin for row of zeros
- construct the auxiliary equation  $A(s) = 0$  by using coefficients of row which is just above the row of zeros
- roots of auxiliary equation are intersection points of root locus with the imaginary axis.

$$s^3 + 5s^2 + 4s + K = 0$$

|       |                                        |                                       |                                   |
|-------|----------------------------------------|---------------------------------------|-----------------------------------|
| $s^3$ | 1                                      | 4                                     |                                   |
| $s^2$ | 5                                      | K                                     | $\Rightarrow A(s) = 5s^2 + K = 0$ |
| $s^1$ | $\frac{5 \cdot 4 - K \cdot 1}{5}$      |                                       |                                   |
| $s^0$ | $\frac{20 - K}{5}$                     | $= 0 \Rightarrow \text{Row of zeros}$ | $\frac{20 - K}{5} = 0$            |
| $s^0$ | $\frac{20 - K}{5} \cdot K - 5 \cdot 0$ |                                       | $20 - K = 0$                      |
|       | $\frac{20 - K}{5}$                     |                                       | $K = 20$                          |
| $s^0$ | K                                      |                                       |                                   |

$$A(s) = 5s^2 + K = 0$$

$$5s^2 + 20 = 0$$

$$s^2 = -20/5$$

$$s^2 = -4$$

$$s = \pm 2j$$

point of intersection  
with imaginary axis

$$s = \sigma + j\omega$$

↓

$$\omega = \pm 2$$

## Procedure for construction of Root Locus

- ↳ locate the poles and zeros of  $G(s)H(s)$  on the s-plane.
- ↳ determine the root locus on real axis.
- ↳ determine the number of branches  $[n-m]$ .
- ↳ determine the meeting point of asymptotes with real axis (centroid).
- ↳ determine the angle of asymptotes with real axis.
- ↳ find the breakaway and breakin points.
- ↳ determine the angle of departure (from complex pole) and angle of arrival (at complex zero).
- ↳ find the crossing point of root loci on imaginary axis.
- ↳ sketch the root locus.

1. A unity feedback system has an open loop transfer function  $G(s)$ . sketch the root locus.

$$G(s) = \frac{k}{s(s^2 + 4s + 13)}$$

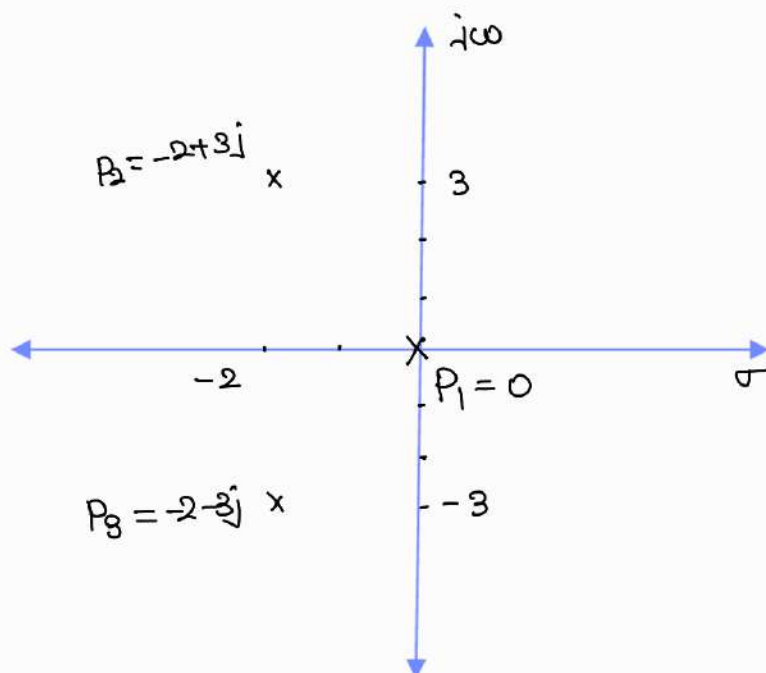
→ location of poles and zeros:

$$G(s)H(s) = \frac{k}{s(s^2 + 4s + 13)} \cdot (1) \quad \left| \quad H(s) = 1 \right.$$

$$= \frac{k}{s(s^2 + 4s + 13)}$$

$$s(s^2 + 4s + 13) = 0 \quad \left| \quad k = 0 \right.$$

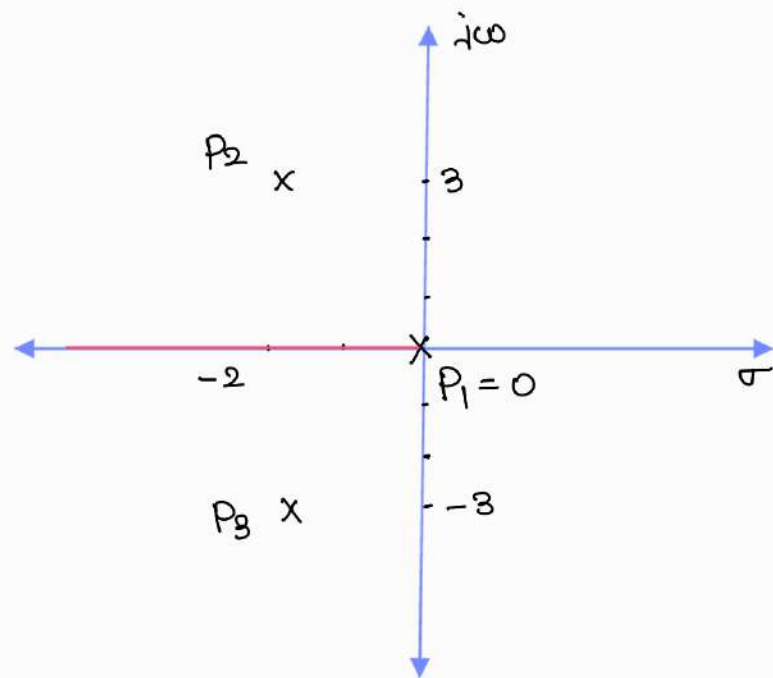
$$s = 0 \quad \left| \quad \begin{aligned} s^2 + 4s + 13 &= 0 \\ s &= \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2(1)} \\ s &= -2 \pm 3j \end{aligned} \right.$$



three poles  
and  
no zeros

→ root locus on real axis:

↳ only one pole on real axis at the origin  
 ⇒ entire negative axis will be part of root locus



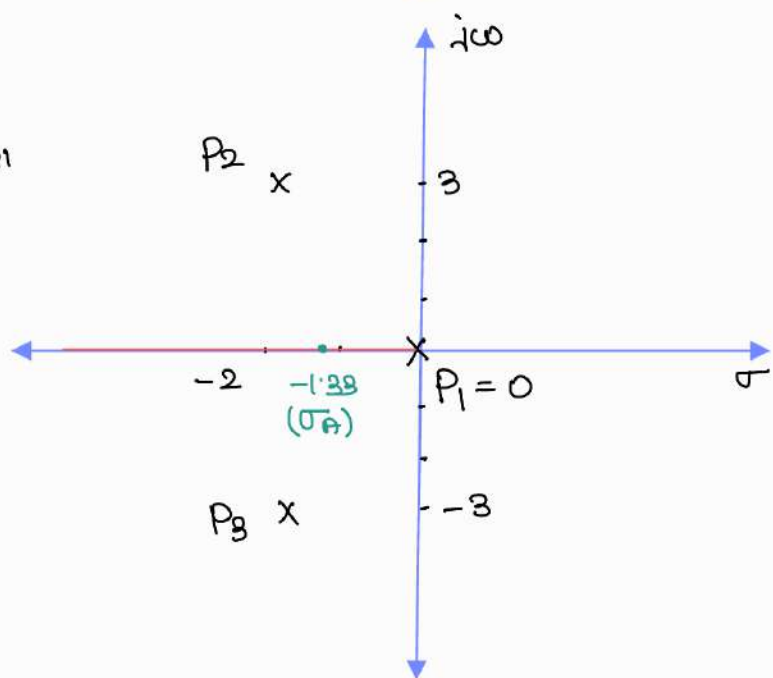
→ centroid:

$$\sigma_A = \frac{\sum P_{\text{real}} - \sum Z_{\text{real}}}{n - m}$$

$$= \frac{0 - 2 - 2 - 0}{3 - 0}$$

$$= -\frac{4}{3}$$

$$\sigma_A = -1.33$$

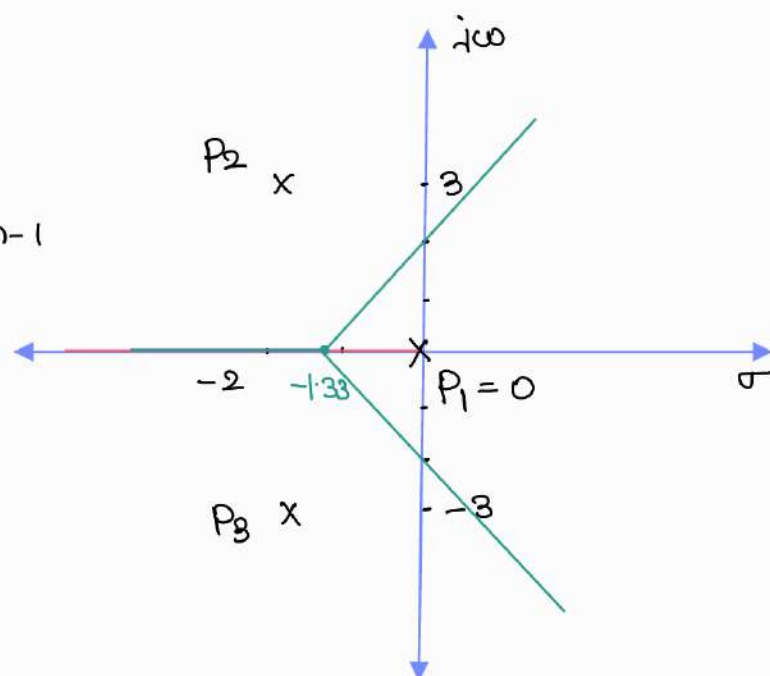


→ angles of asymptotes:

If  $n > m \Rightarrow \theta_A = \frac{(2q+1)180^\circ}{n-m};$

$$q = 0, 1, 2, \dots, n-m-1$$

| $q=0$                            | $q=1$                                     | $q=2$                                     |
|----------------------------------|-------------------------------------------|-------------------------------------------|
| $\theta_A = \frac{180^\circ}{3}$ | $\theta_A = \frac{180^\circ \times 3}{3}$ | $\theta_A = \frac{180^\circ \times 5}{3}$ |
| $\theta_1 = 60^\circ$            | $\theta_2 = 180^\circ$                    | $\theta_3 = 300^\circ$                    |



→ breakaway and breakin points:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{k}{s(s^2 + 4s + 13)}}{1 + \frac{k}{s(s^2 + 4s + 13)}} \quad (1)$$

$$\frac{G(s)}{1 + G(s)H(s)} = \frac{k}{s(s^2 + 4s + 13) + k}$$

$$1 + G(s)H(s) = s(s^2 + 4s + 13) + k = 0$$

$$k = -(s)(s^2 + 4s + 13)$$

$$k = -s^3 - 4s^2 - 13s$$

$$\frac{dk}{ds} = 0$$

$$\frac{d}{ds}(-s^3 - 4s^2 - 13s) = 0$$

$$-3s^2 - 8s - 13 = 0$$

$$3s^2 + 8s + 13 = 0$$

$$s = \frac{-8 \pm \sqrt{8^2 - 4(3)(13)}}{2(3)}$$

$$s = -1.33 \pm j1.6$$

$$k = -(s^3 + 4s^2 + 13s)$$

$$= -[(-1.33 \pm j1.6)^3 + 4(-1.33 \pm j1.6)^2 + 13(-1.33 \pm j1.6)]$$

$k \neq$  positive and real

↳ root locus has neither breakaway nor breakin point.

→ angle of departure and angle of arrival:

$$\phi_d = 180^\circ - [\sum \phi_p + \sum \phi_z]$$

$$\phi_1 = 180^\circ - \tan^{-1}\left(\frac{3}{2}\right)$$

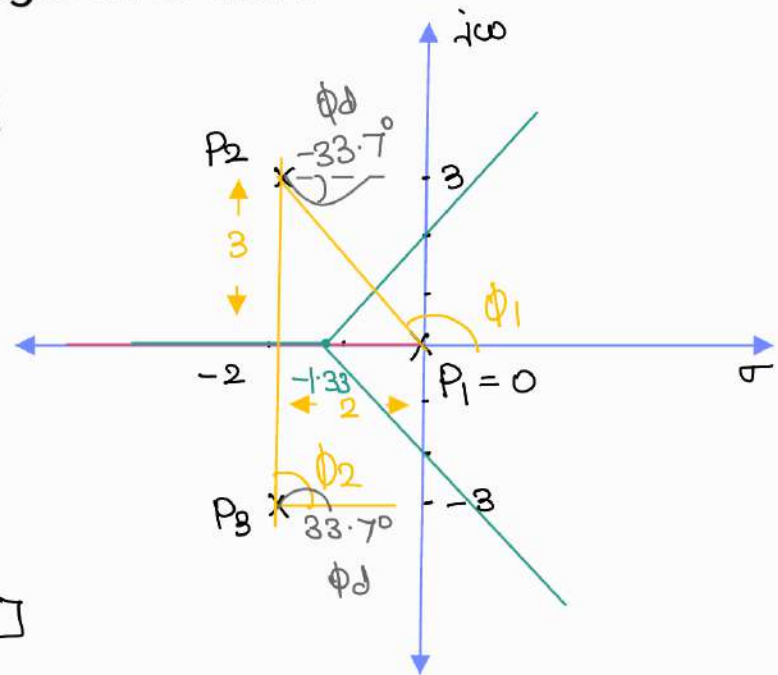
$$\phi_1 = 123.7^\circ$$

$$\phi_2 = 90^\circ$$

$$\phi_d = 180^\circ - [123.7^\circ + 90^\circ]$$

$$\phi_d = -33.7^\circ \quad \text{at } P_2$$

$$\phi_d = 33.7^\circ \quad \text{at } P_3$$

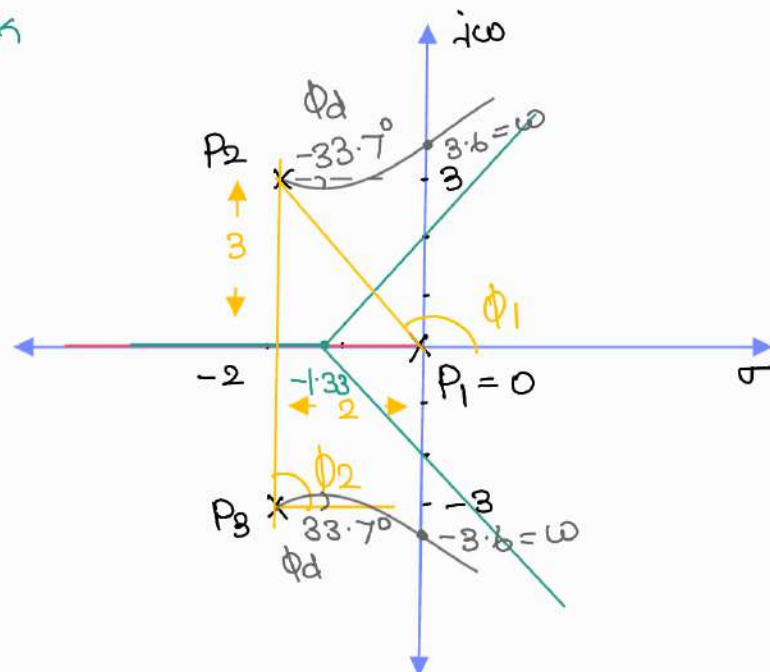


→ crossing point on imaginary axis:

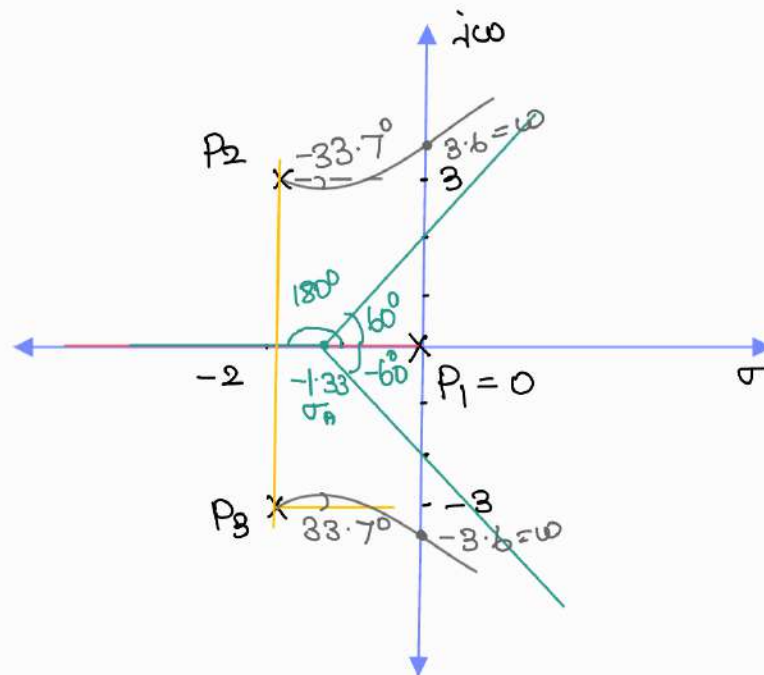
$$s^3 + 4s^2 + 13s + K = 0$$

|       |                                                     |                                 |
|-------|-----------------------------------------------------|---------------------------------|
| $s^3$ | 1                                                   | 13                              |
| $s^2$ | 4                                                   | $K \Rightarrow A(s) = 4s^2 + K$ |
| $s^1$ | $\frac{4 \times 13 - 1 \times K}{4}$                |                                 |
| $s^0$ | $\frac{52 - K}{4} \Rightarrow \text{Row of zeros:}$ |                                 |
| $s^0$ | $\frac{52 - K \times K}{4} = 0$                     | $\frac{52 - K}{4} = 0$          |
| $s^0$ | $K$                                                 | $K = 52$                        |

$4s^2 + K = 0$   
 $4s^2 + 52 = 0$   
 $s^2 = -\frac{52}{4}$   
 $s = \pm j3.6$   
 $\omega = \pm 3.6$



→ root locus:



→ root locus branches:

- ↳ one starts at origin and travel through negative axis (parallel to asymptote) to meet the zero at infinity.
- ↳ two other starts at complex poles with angle of departure  $\pm 33.7^\circ$ , crossing the imaginary axis at  $\pm 3.6j$  and travel parallel to asymptotes to meet the zeros at infinity.
- ↳ limiting value of K for the stability of the system is 52.

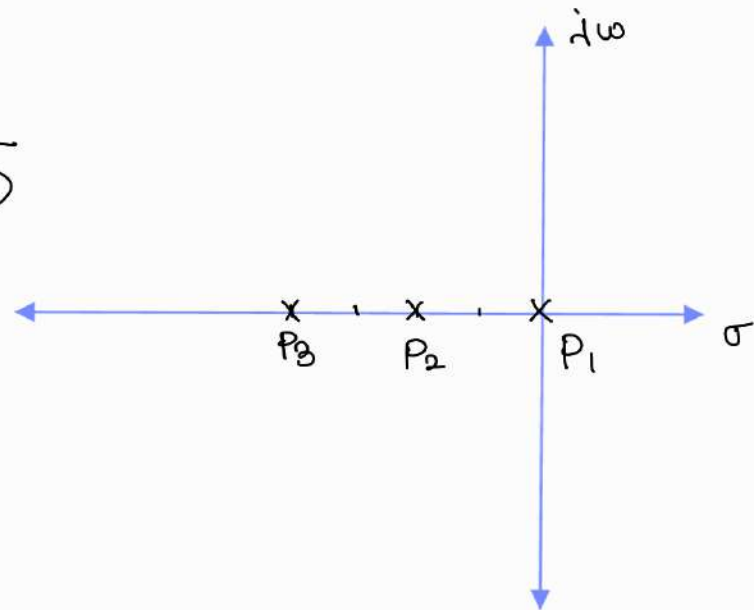
2. sketch the root locus of the system whose open loop transfer function is  $G(s)$ .

$$G(s) = \frac{k}{s(s+2)(s+4)}$$

→ location of poles and zeros:

$$G(s)H(s) = \frac{k}{s(s+2)(s+4)}$$

$$\begin{array}{l|l} s(s+2)(s+4) = 0 & k=0 \\ s = 0, -2, -4 & \end{array}$$



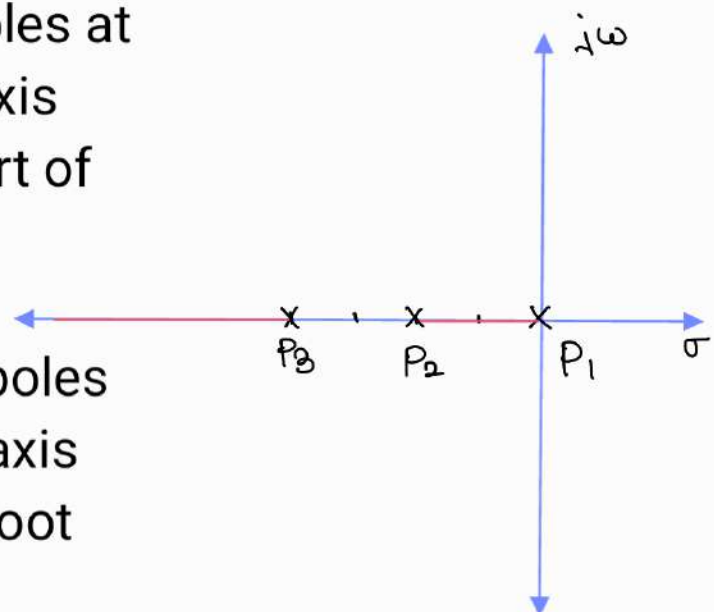
↳ three poles and no zeros

→ root locus on real axis:

↳ between (0, -2) ⇒ one pole at right side ⇒ odd ⇒ real axis  
between (0, -2) ⇒ part of root locus.

↳ between (-2, -4) ⇒ two poles at right side ⇒ even ⇒ real axis  
between (-2, -4) ⇒ not a part of root locus.

↳ between (-4,  $-\infty$ ) ⇒ three poles at right side ⇒ odd ⇒ real axis  
between (-4,  $-\infty$ ) ⇒ part of root locus.



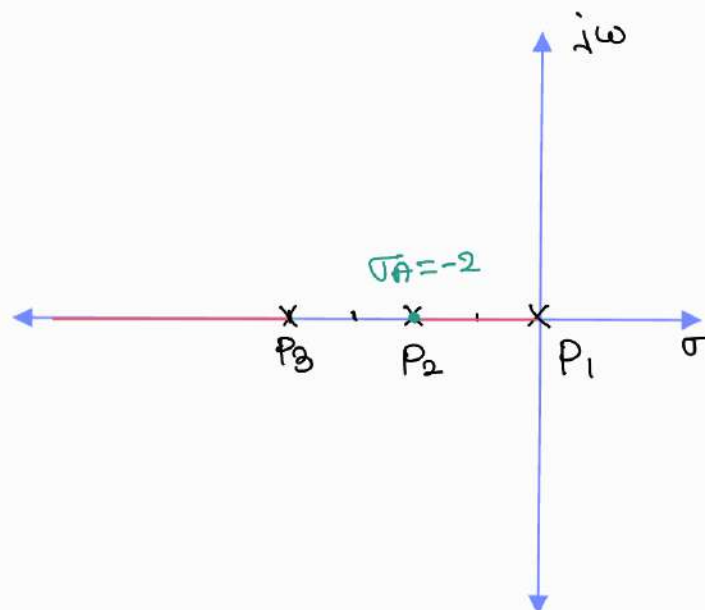
→ centroid:

$$\sigma_A = \frac{\sum P_{\text{real}} - \sum Z_{\text{real}}}{n - m}$$

$$= \frac{(0 - 2 - 4) - 0}{3 - 0}$$

$$= \frac{-6}{3}$$

$$\sigma_A = -2$$

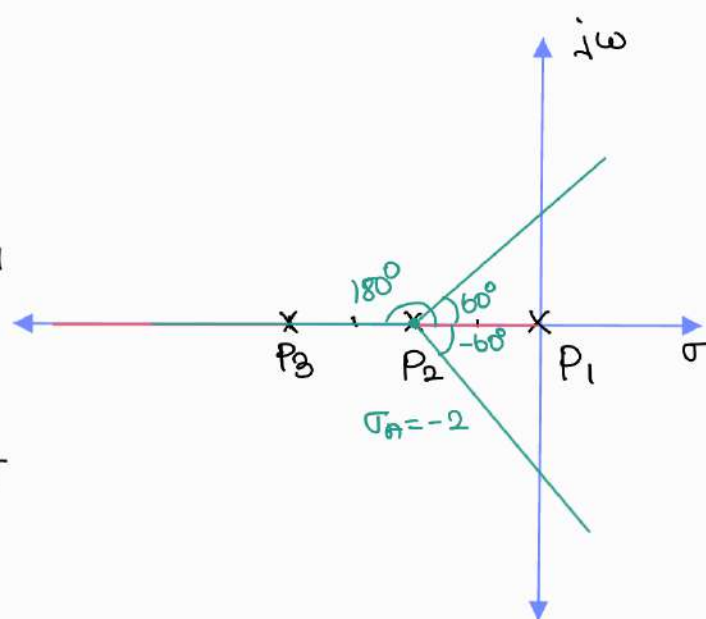


→ angles of asymptotes:

$$\text{If } n > m \Rightarrow \theta_A = \frac{(2q+1)180^\circ}{n-m};$$

$$q = 0, 1, 2, \dots, n-m-1$$

|                                  |                                           |                                           |
|----------------------------------|-------------------------------------------|-------------------------------------------|
| $q=0$                            | $q=1$                                     | $q=2$                                     |
| $\theta_A = \frac{180^\circ}{3}$ | $\theta_A = \frac{180^\circ \times 3}{3}$ | $\theta_A = \frac{180^\circ \times 5}{3}$ |
| $\theta_1 = 60^\circ$            | $\theta_2 = 180^\circ$                    | $\theta_3 = 300^\circ$                    |



→ breakaway and breakin points:

$$T(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{k}{s(s+2)(s+4)}}{1 + \frac{k}{s(s+2)(s+4)}} = \frac{k}{s(s+2)(s+4) + k}$$

$$1+G(s)H(s) = s(s+2)(s+4) + k = 0$$

$$k = -s(s+2)(s+4) = -s(s^2 + 6s + 8)$$

$$k = -s^3 - 6s^2 - 8s$$

$$\frac{dk}{ds} = -3s^2 - 12s - 8 < 0$$

$$3s^2 + 12s + 8 = 0$$

$$s = \frac{-12 \pm \sqrt{12^2 - 4(3)(8)}}{2(3)}$$

$$s = -0.845 \text{ (or)} -3.154$$

$$s = -0.845$$

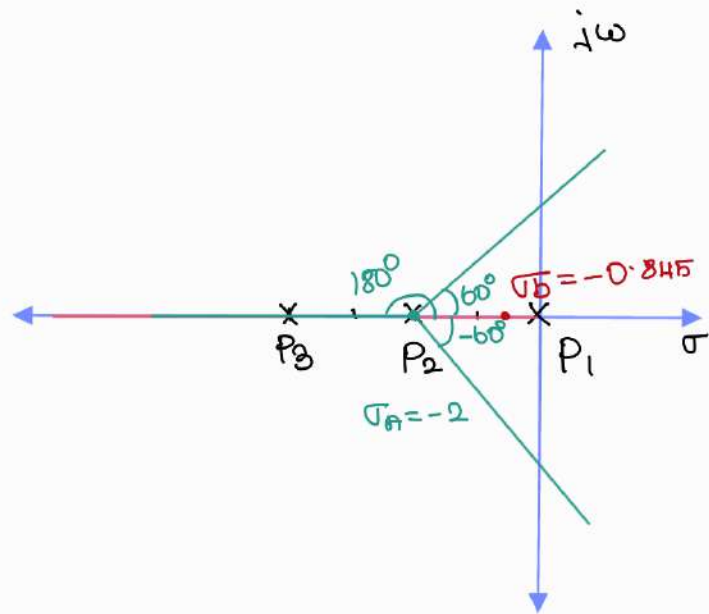
$$k = 3.08$$

positive and real

$$s = -3.154$$

$$k = -3.08$$

negative



$$\sigma_B = -0.845$$

→ angle of departure and angle of arrival:

↳ no complex pole and zero ⇒ neither angle of departure nor angle of arrival.

→ crossing point on imaginary axis:

$$s^3 + 6s^2 + 8s + k = 0$$

|       |                  |                                                  |
|-------|------------------|--------------------------------------------------|
| $s^3$ | 1                | 8                                                |
| $s^2$ | 6                | $k \Rightarrow$                                  |
| $s^1$ | $\frac{48-k}{6}$ | $\Rightarrow$ Row of zeros: $\frac{48-k}{6} = 0$ |
| $s^0$ | $k$              | $48-k=0$<br>$k=48$                               |

$$A(s) = 6s^2 + k$$

$$6s^2 + 48 = 0$$

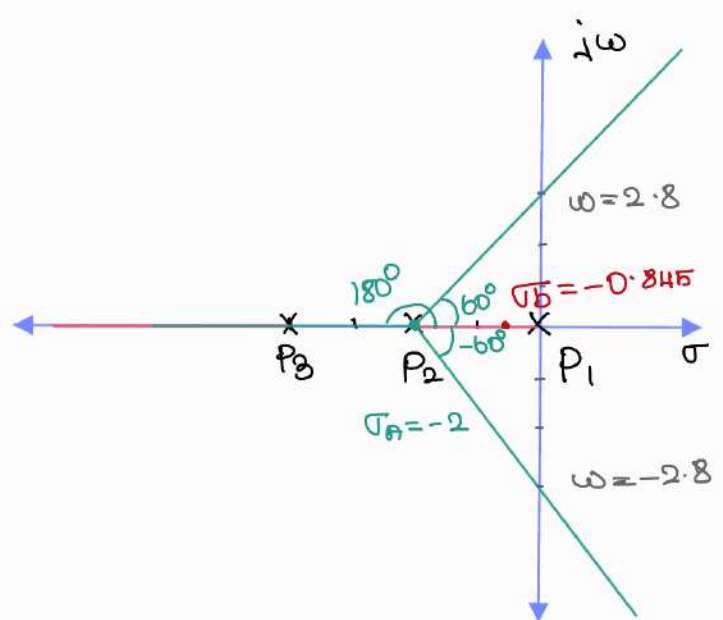
$$6s^2 + 48 = 0$$

$$6s^2 = -48$$

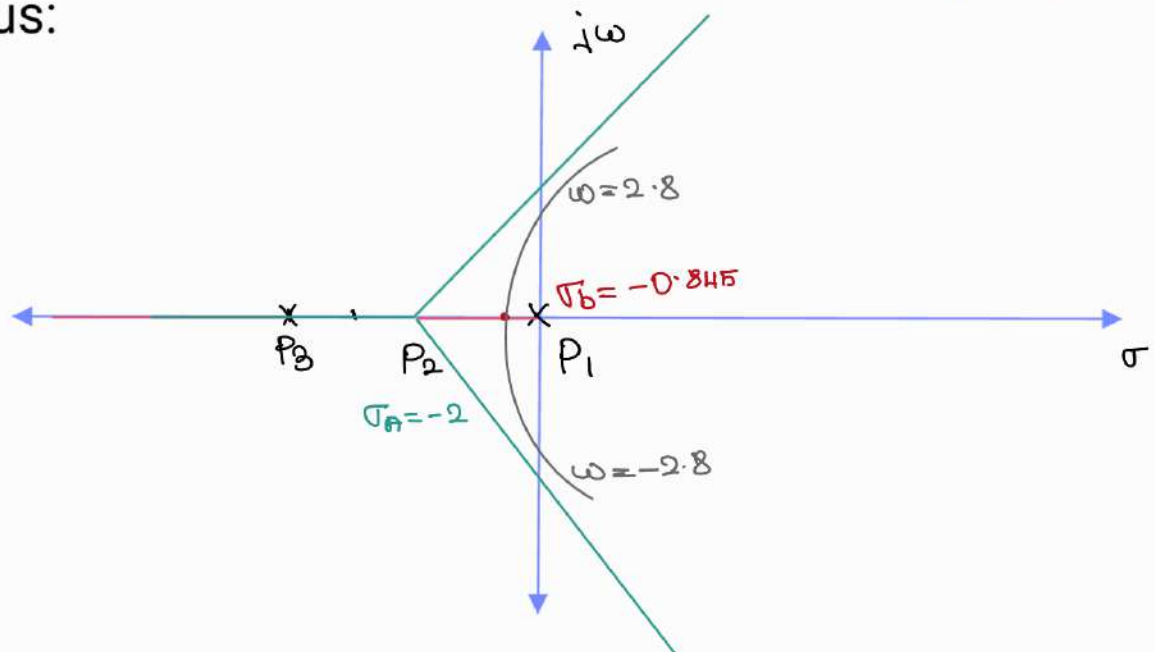
$$s^2 = -8$$

$$s = \pm 2.8j$$

$$\omega = \pm 2.8$$



→ root locus:



→ root locus branches:

- ↳ one starts at  $s=-4$  and travel through negative axis (parallel to asymptote) to meet the zero at infinity.
- ↳ two other starts at  $s=0$  and  $s=-2$  with breakaway point on real axis  $-0.845$ , crossing the imaginary axis at  $\pm 2.8j$  and travel parallel to asymptotes to meet the zeros at infinity.
- ↳ limiting value of  $K$  for the stability of the system is  $48$ .

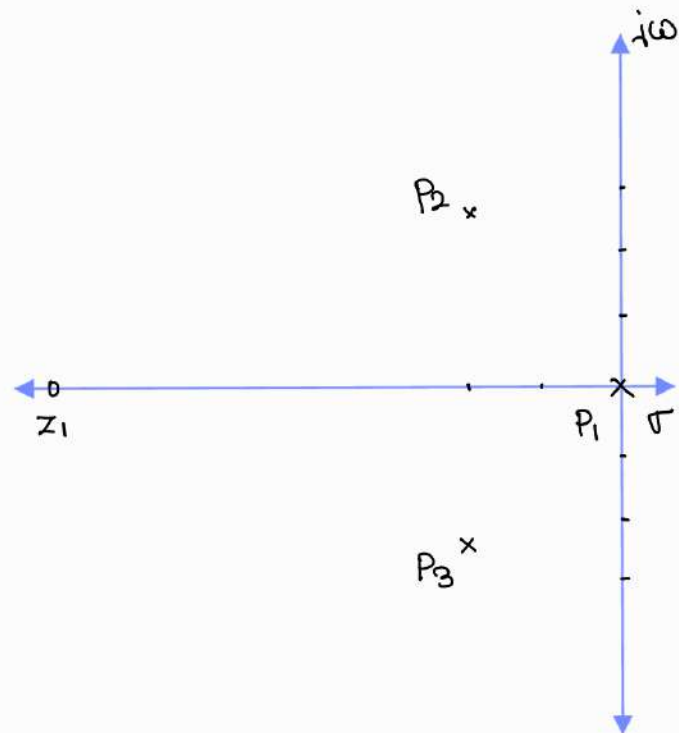
3. A unity feedback system has an open loop transfer function  $G(s)$ . sketch the root locus.

$$G(s) = \frac{k(s+9)}{s(s^2+4s+11)}$$

→ location of poles and zeros:

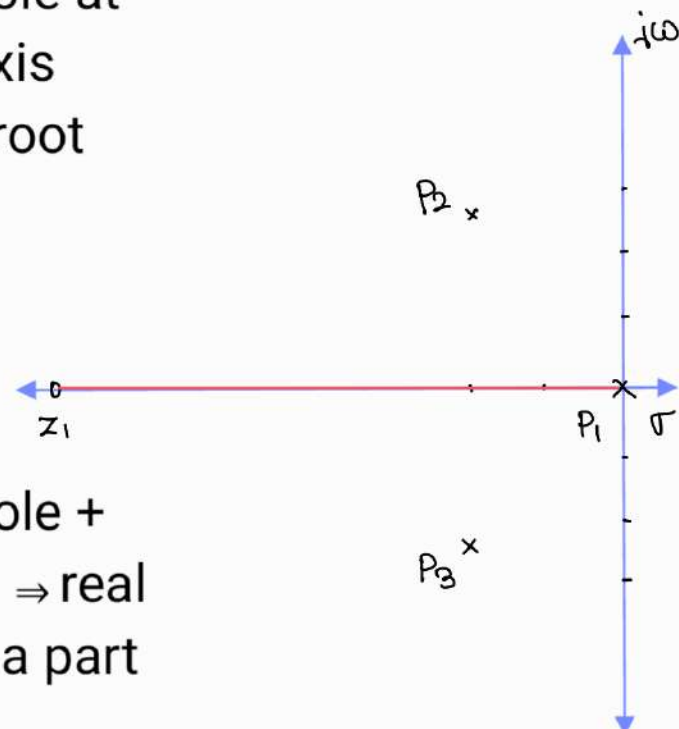
$$G(s)H(s) = \frac{k(s+9)}{s(s^2+4s+11)}$$

$$\begin{array}{l|l|l} s+9=0 & s=0 & s^2+4s+11=0 \\ s=-9 & P_1=0 & s = \frac{-4 \pm \sqrt{4^2 - 4(11)(1)}}{2(1)} \\ Z_1=-9 & & s = -2 \pm j2.64 \\ & & P_2 = -2 + j2.64 ; P_3 = -2 - j2.64 \end{array}$$



→ root locus on real axis:

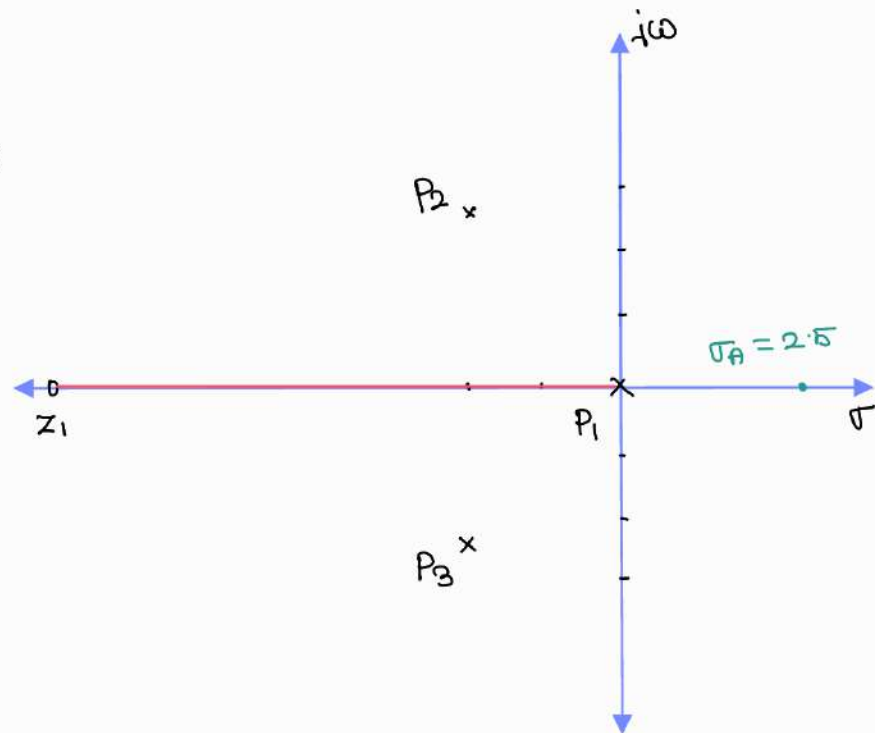
↳ between  $(0, -9) \Rightarrow$  one pole at right side  $\Rightarrow$  odd  $\Rightarrow$  real axis between  $(0, -9) \Rightarrow$  part of root locus.



↳ between  $(-9, \infty) \Rightarrow$  two (pole + zero) at right side  $\Rightarrow$  even  $\Rightarrow$  real axis between  $(-9, \infty) \Rightarrow$  not a part of root locus.

→ centroid:

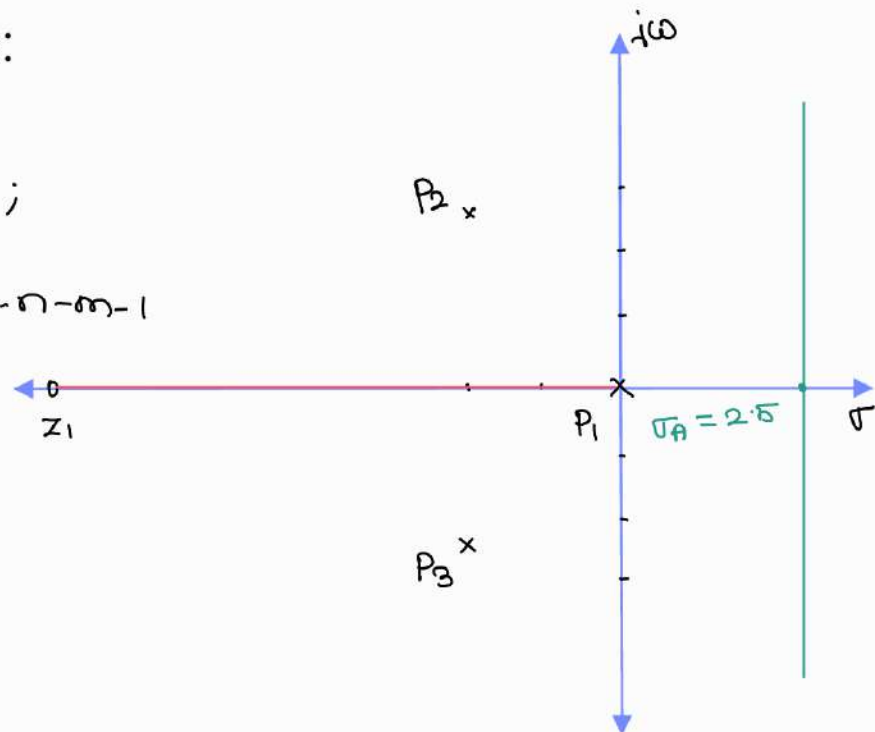
$$\begin{aligned}\sigma_A &= \frac{\sum P_{\text{real}} - \sum Z_{\text{real}}}{n-m} \\ &= \frac{(0-2-2) - (-9)}{3-1} \\ &= \frac{5}{2} \\ \sigma_A &= 2.5\end{aligned}$$



→ angles of asymptotes:

If  $n > m \Rightarrow \theta_A = \frac{(2q+1)180^\circ}{n-m};$   
 $q = 0, 1, 2, \dots, n-m-1$

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| $q=0$                            | $q=1$                                     |
| $\theta_A = \frac{180^\circ}{2}$ | $\theta_A = \frac{180^\circ \times 3}{2}$ |
| $\theta_1 = 90^\circ$            | $\theta_2 = 270^\circ$                    |



→ breakaway and breakin points:

$$T(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{k(s+9)}{s(s^2+4s+11)}}{1 + \frac{k(s+9)}{s(s^2+4s+11)}} = \frac{k(s+9)}{s(s^2+4s+11) + k(s+9)}$$

$$1+G(s)H(s) = s(s^2+4s+11) + k(s+9) = 0$$

$$s^3 + 4s^2 + 11s + k(s+9) = 0$$

$$K(s+9) = -(s^3 + 4s^2 + 11s)$$

$$K = - \frac{(s^3 + 4s^2 + 11s)}{(s+9)}$$

$$\frac{dK}{ds} = - \left[ \frac{(3s^2 + 8s + 11)(s+9) - (s^3 + 4s^2 + 11s)(1)}{(s+9)^2} \right] = 0$$

$$-3s^3 - 8s^2 - 11s - 27s^2 - 72s - 99 + s^3 + 4s^2 + 11s = 0$$

$$-2s^3 - 31s^2 - 72s - 99 = 0$$

$$2s^3 + 31s^2 + 72s + 99 = 0$$

$$(s+3)(2s+3)(s+11) = 0$$

$$s = -3$$

$$K = - \frac{(s^3 + 4s^2 + 11s)}{(s+9)}$$

$$= - \frac{(-27 + 36 - 33)}{6}$$

$$K = 4$$

$$\sigma_b = -3$$

$$s = -\frac{3}{2}$$

$$K = - \frac{(s^3 + 4s^2 + 11s)}{(s+9)}$$

$$= - \left[ \frac{-\frac{27}{8} + 9 - \frac{33}{2}}{-\frac{3}{2} + 9} \right]$$

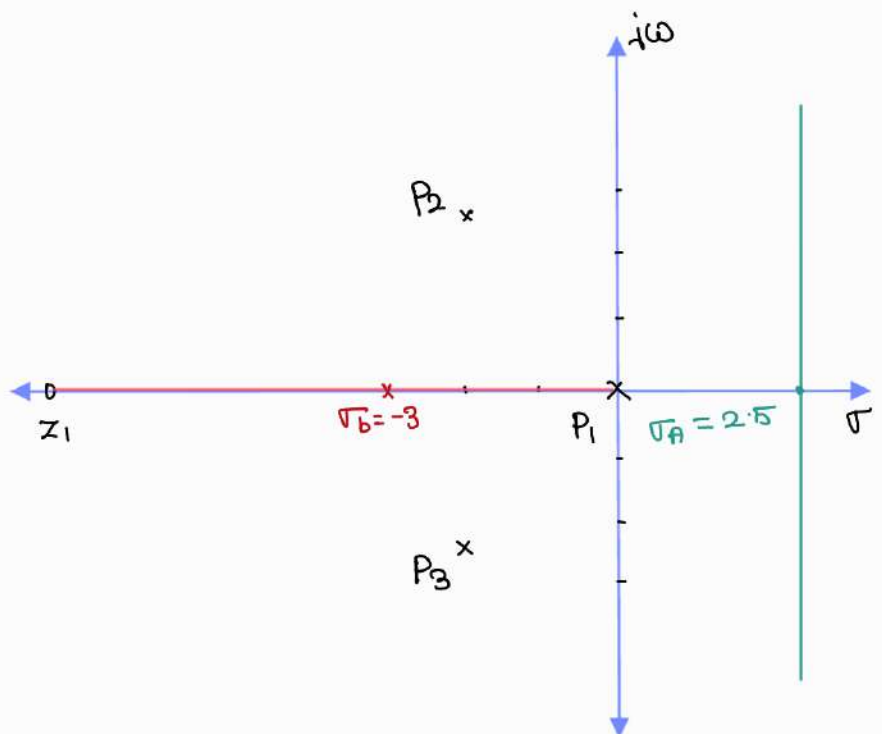
$$K = \left[ -\frac{87}{15} \right] \Rightarrow \text{negative}$$

$$s = -11$$

$$K = - \frac{(s^3 + 4s^2 + 11s)}{(s+9)}$$

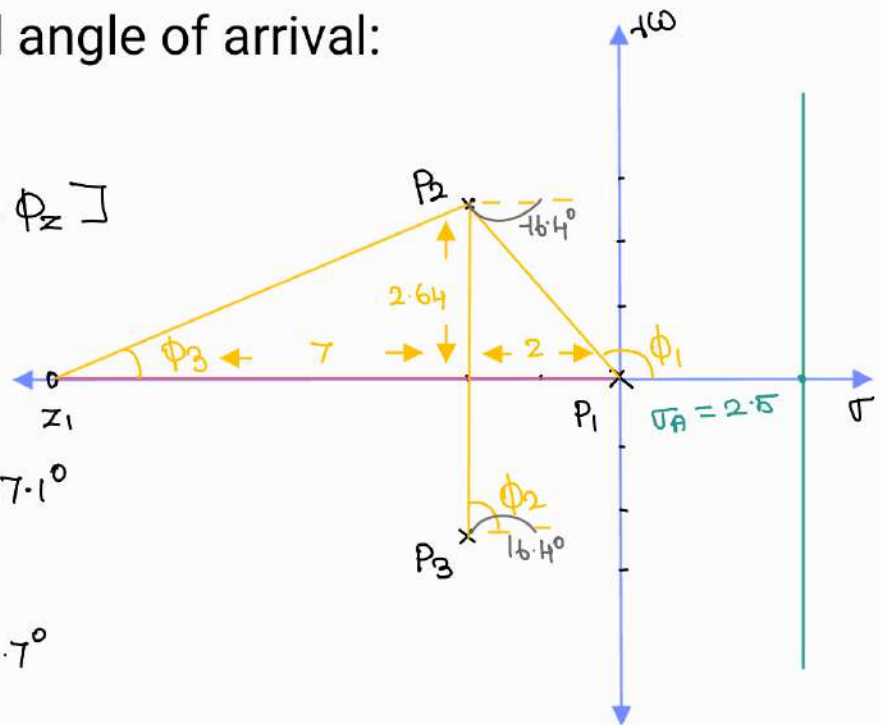
$$= - \frac{[-1331 + 484 - 121]}{-2}$$

$$K = -484 \Rightarrow \text{negative}$$



→ angle of departure and angle of arrival:

$$\phi_d = 180^\circ - [\angle \phi_p + \angle \phi_z]$$



$$\phi_1 = 180^\circ - \tan^{-1}\left(\frac{2.64}{2}\right) = 127.1^\circ$$

$$\phi_2 = 90^\circ$$

$$\phi_3 = \tan^{-1}\left(\frac{2.64}{7}\right) = 20.7^\circ$$

$$\phi_d = 180^\circ - [127.1^\circ + 90^\circ] + [20.7^\circ]$$

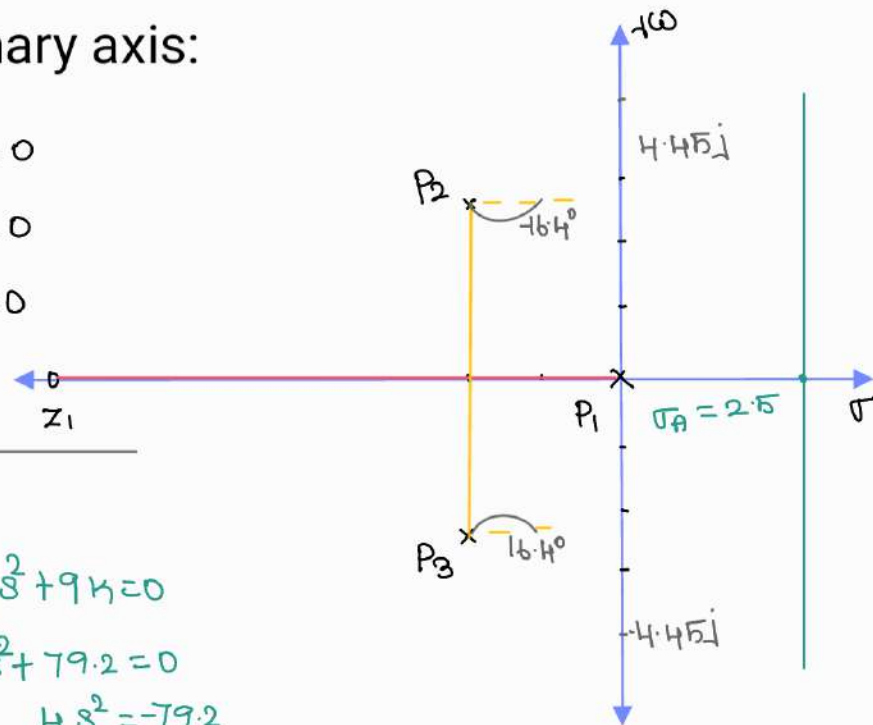
$$\phi_d = -16.4^\circ \text{ at } P_2 \Rightarrow \phi_d = +16.4^\circ \text{ at } P_1$$

→ crossing point on imaginary axis:

$$s^3 + 4s^2 + 11s + k(s+9) = 0$$

$$s^3 + 4s^2 + 11s + ks + 9k = 0$$

$$s^3 + 4s^2 + (11+k)s + 9k = 0$$



$$s^3 \quad 1 \quad 11+k$$

$$s^2 \quad 4 \quad 9k \Rightarrow R(s) = 4s^2 + 9k = 0$$

$$4s^2 + 79.2 = 0$$

$$4s^2 = -79.2$$

$$s^2 = \frac{-79.2}{4}$$

$$s^1 \quad \frac{44+4k-9k}{4} \Rightarrow \text{Row of zeros: } s = \pm j4.45$$

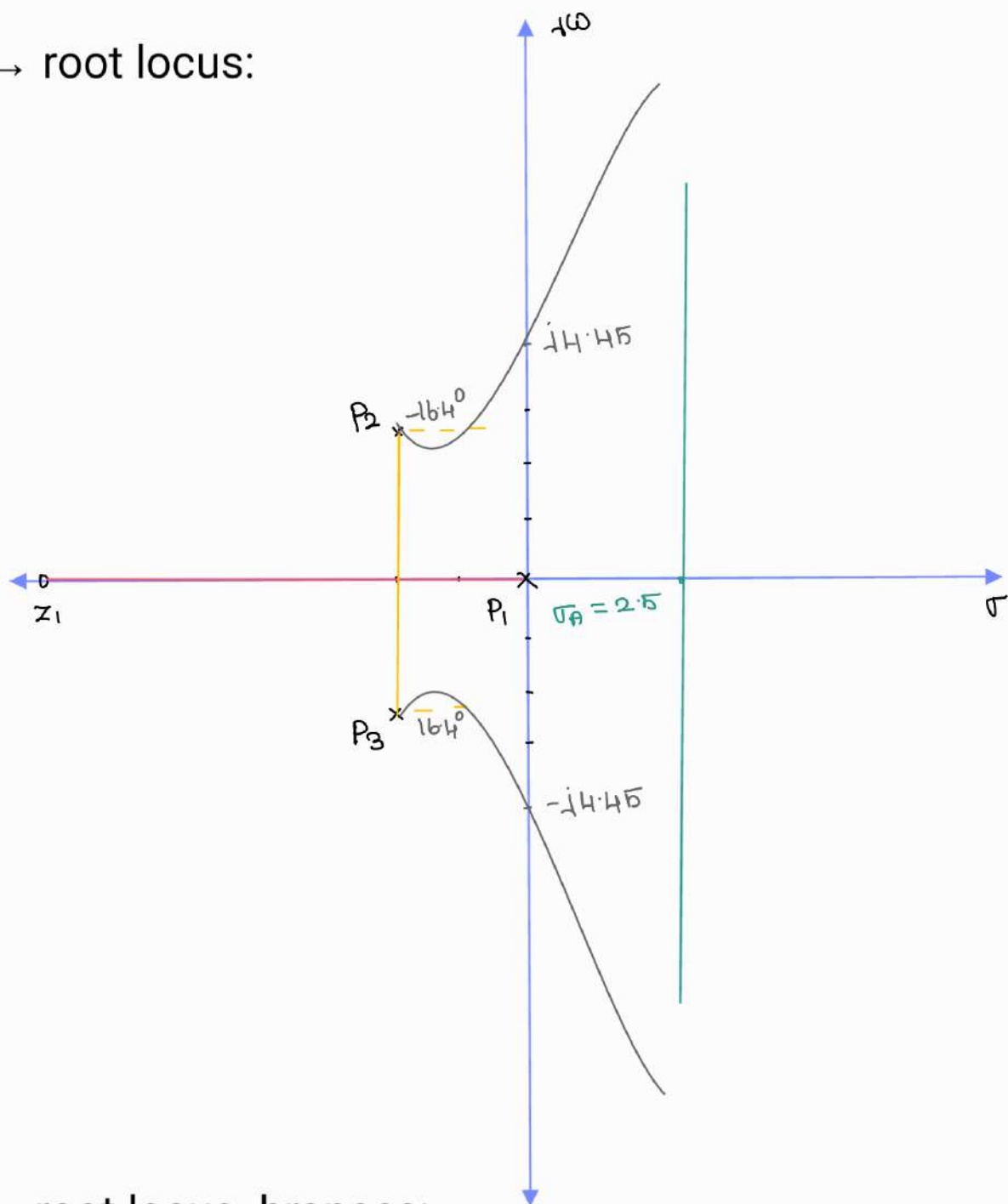
$$s^1 \quad \frac{44-5k}{4} \Rightarrow \frac{44-5k}{4} = 0$$

$$s^0 \quad \frac{44-5k \times 9k - 0}{4} \Rightarrow 44 = 5k$$

$$s^0 \quad 9k \Rightarrow k = \frac{44}{5} = 8.8$$

$$\omega = \pm 4.45$$

→ root locus:



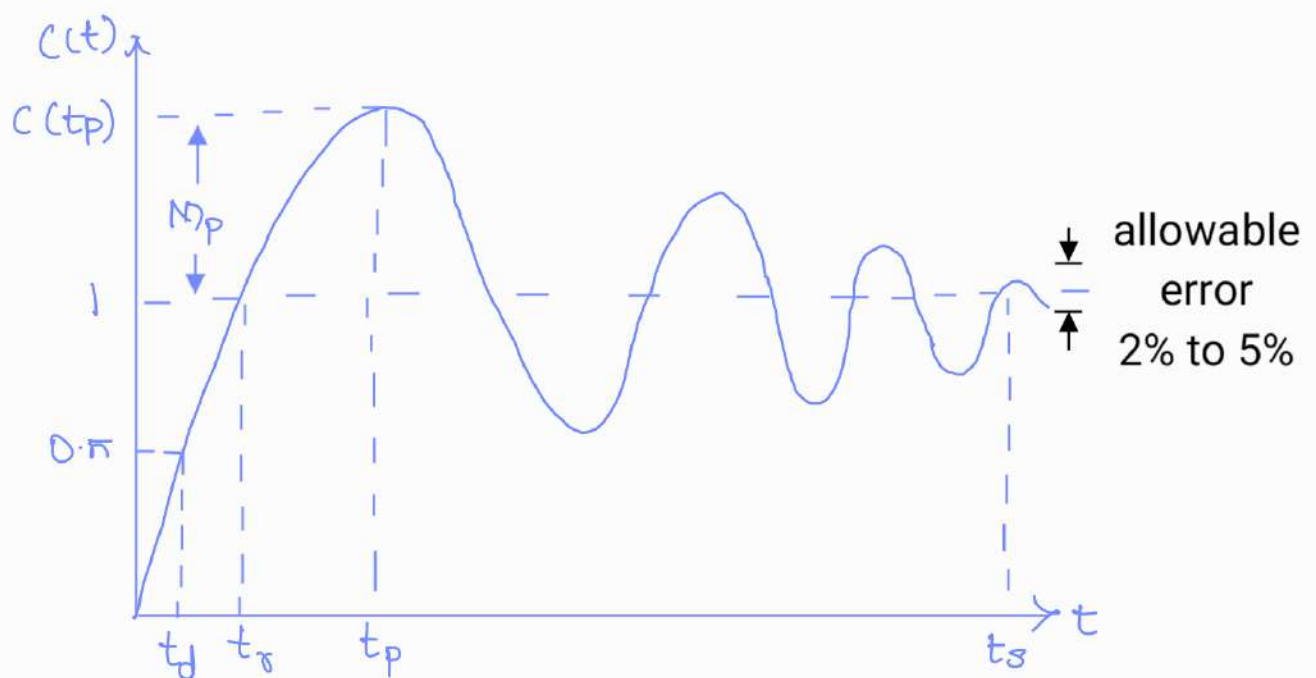
→ root locus branches:

- ↳ one starts at  $s=0$  and travel through negative axis to meet the zero at  $s=-9$ .
- ↳ two other starts at complex poles with angle of departure  $\pm 16.4^\circ$ , crossing the imaginary axis at  $\pm 4.45j$  and travel parallel to asymptotes to meet the zeros at infinity.
- ↳ limiting value of  $K$  for the stability of the system is 8.8.

# Time Domain Specifications

→ Desired performance characteristics of control systems are specified in terms of time domain specifications.

- ↳ delay time
- ↳ rise time
- ↳ peak time
- ↳ maximum overshoot
- ↳ settling time



→ **Delay time:** time taken for response to reach 50% of the final value, for the very first time.

→ **Rise time:** time taken for response to raise from 0 to 100% for the very first time.

- ⇒ under damped: 0 to 100%
- ⇒ critically damped: 5 to 95%
- ⇒ over damped: 10 to 90%

→ **Peak time:** time taken for response to reach the peak value for the very first time.

→ **Peak Overshoot:** ratio of the maximum peak value to the final value, where the maximum peak value is measured from final value.

$$M_p \%. = \frac{c(t_p) - c(t_\infty)}{c(t_\infty)} \times 100$$

→ **Settling time:** time taken by the response to reach and stay within a specified error.

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$\theta = \tan^{-1} \frac{\sqrt{1 - \zeta^2}}{\zeta}$$

$$t_p = \frac{\pi}{\omega_d}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\% \text{ of } M_p = e^{-\frac{\zeta \pi}{\sqrt{1 - \zeta^2}}} \times 100$$

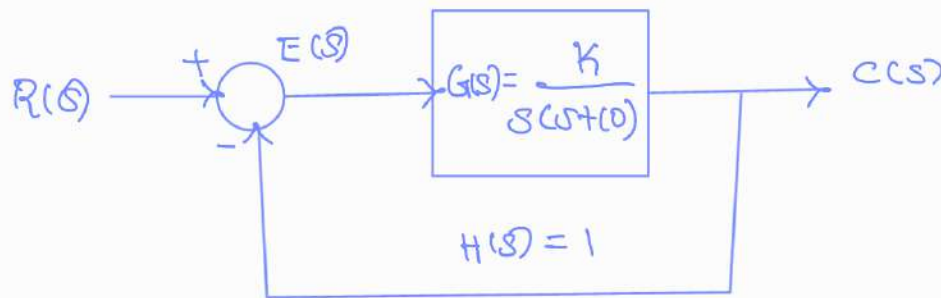
$$T = \frac{1}{\zeta \omega_n}$$

$$t_s = \frac{4}{\zeta \omega_n} = 4T \quad \text{for } 2\% \text{ of error}$$

$$t_s = \frac{3}{\zeta \omega_n} = 3T \quad \text{for } 5\% \text{ of error}$$

$$t_s = \frac{\ln(\% \text{ error})}{\zeta \omega_n} = \frac{\ln(\% \text{ error})}{T}$$

1. The unity feedback system is characterized by an open loop transfer function  $G(s)$ . Determine the gain  $K$ , so that the system will have a damping ratio of 0.5 for this value of  $K$ . Determine peak overshoot and time at overshoot for a unit step input.



closed loop transfer function:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{K/s(s+10)}{1 + \left[ \frac{K}{s(s+10)} \cdot 1 \right]}$$

$$= \frac{K/s(s+10)}{s^2 + 10s + K/s(s+10)}$$

$$\frac{C(s)}{R(s)} = \frac{K}{s^2 + 10s + K} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n^2 = K$$

$$\omega_n = \sqrt{K}$$

$$2\zeta\omega_n = 10$$

$$2 \times (0.5) \omega_n = 10$$

$$\zeta = 0.5$$

$$\sqrt{K} = 10$$

$$\text{Gain: } K = 100$$

$$\text{Natural frequency of oscillation: } \omega_n = 10 \text{ rad/sec}$$

$$\% \text{ of } M_p = e^{-\frac{\zeta \pi}{\sqrt{1-\zeta^2}}} \times 100$$

$$= e^{-\frac{0.5 \pi}{\sqrt{1-0.5^2}}} \times 100$$

$$= e^{-\frac{1.571}{0.866}} \times 100$$

$$= e^{-1.814} \times 100$$

$$\% \text{ of } M_p = 0.163 \times 100$$

$$\text{Peak overshoot} = 16.3 \%$$

$$t_p = \frac{\pi}{\omega_d} \quad \omega_d = \omega_n \sqrt{1-\zeta^2}$$

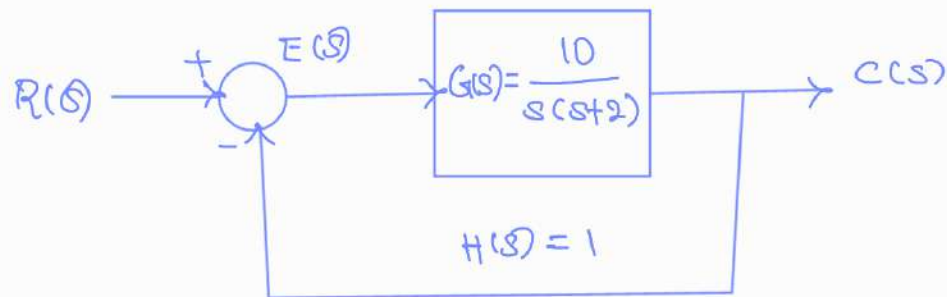
$$= \frac{\pi}{10 \sqrt{1-0.5^2}}$$

$$= \frac{\pi}{10 \times 0.866}$$

$$= \frac{\pi}{8.66}$$

$$\text{Peak time} = 0.363 \text{ sec}$$

2. A unity feedback control system has an open loop transfer function  $G(s)$ . Find the rise time, percentage overshoot, peak time and settling time for a step input of 12 units.



closed loop transfer function:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{10 / s(s+2)}{1 + \left[ \frac{10}{s(s+2)} \cdot 1 \right]}$$

$$= \frac{10 / s(s+2)}{s^2 + 2s + 10 / s(s+2)}$$

$$\frac{C(s)}{R(s)} = \frac{10}{s^2 + 2s + 10} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n^2 = 10$$

$$\omega_n = \sqrt{10} = 3.162 \text{ rad/sec}$$

$$2\zeta\omega_n = 2$$

$$\zeta = \frac{2}{2} \times \frac{1}{\sqrt{10}}$$

$$\zeta = 0.316$$

$$\begin{aligned}
 \theta &= \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} \\
 &= \tan^{-1} \frac{\sqrt{1-0.316^2}}{0.316} \\
 &= \tan^{-1} \frac{0.949}{0.316} \\
 &= \tan^{-1} 3.002
 \end{aligned}$$

$$\theta = 1.249 \text{ rad}$$

$$\begin{aligned}
 \omega_d &= \omega_n \sqrt{1-\zeta^2} \\
 &= 3.162 \times \sqrt{1-0.316^2} \\
 &= 3.162 \times 0.949 \\
 \omega_d &= 3 \text{ rad/sec}
 \end{aligned}$$

Rise time:

$$\begin{aligned}
 t_r &= \frac{\pi - \theta}{\omega_d} \\
 &= \frac{\pi - 1.249}{3} \\
 t_r &= 0.63 \text{ sec}
 \end{aligned}$$

Percentage overshoot:  $\%M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100$

$$\begin{aligned}
 &= e^{\frac{-0.316\pi}{\sqrt{1-0.316^2}}} \times 100 \\
 &= e^{-0.993/0.949} \times 100 \\
 &= e^{-1.046} \times 100 \\
 &= 0.351 \times 100
 \end{aligned}$$

$$\%M_p = 35.1 \%$$

Peak overshoot:

$$M_p = \frac{35.1}{100} \times 12$$

$$M_p = 4.212 \text{ units}$$

Peak time:

$$t_p = \frac{\pi}{\omega_d}$$

$$= \frac{\pi}{3}$$

$$t_p = 1.047 \text{ sec}$$

Time constant:

$$T = \frac{1}{\zeta \omega_n}$$

$$= \frac{1}{0.316 \times 3.162}$$

$$T = 1 \text{ sec}$$

Settling time: for 2% of error =  $4T = 4 \text{ sec}$   
for 5% of error =  $3T = 3 \text{ sec}$

3. A closed loop servo is represented by the differential equation, where  $c$  is the displacement of the output shaft,  $r$  is the displacement of the input shaft and  $e = r - c$ . Determine undamped natural frequency, damping ratio and percentage maximum overshoot for unit step input.

$$\frac{d^2 c}{dt^2} + 8 \frac{dc}{dt} = 64 e$$

$$\frac{d^2 c}{dt^2} + 8 \frac{dc}{dt} = 64e$$

$$\frac{d^2 c}{dt^2} + 8 \frac{dc}{dt} = 64 (r - c)$$

Laplace transform:

$$s^2 C(s) + 8s C(s) = 64 [R(s) - C(s)]$$

$$s^2 C(s) + 8s C(s) + 64 C(s) = 64 R(s)$$

$$C(s) [s^2 + 8s + 64] = 64 R(s)$$

$$\frac{C(s)}{R(s)} = \frac{64}{s^2 + 8s + 64} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n^2 = 64$$

undamped natural frequency:  $\omega_n = 8 \text{ rad/sec}$

$$2\zeta\omega_n = 8$$

$$\zeta = \frac{8}{2 \times 8}$$

damping ratio:

$$\zeta = 0.5$$

$$\begin{aligned} \%M_p &= e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 \\ &= e^{\frac{-0.5\pi}{\sqrt{1-0.5^2}}} \times 100 \\ &= e^{-1.57/0.866} \times 100 \\ &= e^{-1.813} \times 100 \end{aligned}$$

percentage maximum overshoot:  $\%M_p = 16.31 \%$

## Type Number of Control Systems

- Specified for loop transfer function  $[G(s).H(s)]$ .
- Number of poles of the loop transfer function lying at origin decides the number of control system.
- if  $N$  is number of poles at origin then Type number is  $N$

$$G(s)H(s) = K \frac{C(s)}{R(s)} = K \frac{(s+z_1)(s+z_2)\dots(s+z_m)}{s^N (s+p_1)(s+p_2)\dots(s+p_n)}$$

where  $z_1, z_2, \dots, z_m \rightarrow \text{zeros}$

$p_1, p_2, \dots, p_n \rightarrow \text{poles}$

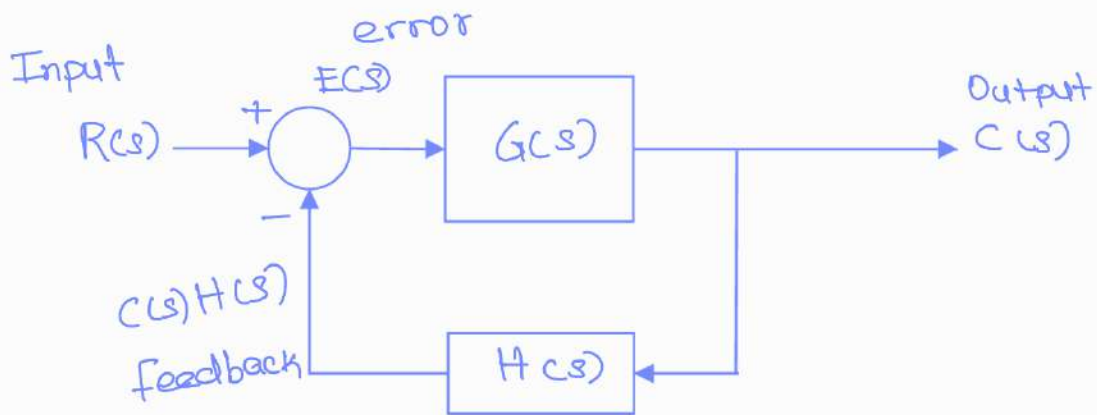
$K \rightarrow \text{Constant}$

$N \rightarrow \text{number of poles at origin}$

$$G(s)H(s) = \frac{K(s+2)}{s(s^2+2s+3)}$$

Type number of the system = 1 [number of poles at origin]  
Order of the system = 3 [total number of poles]

## Error Signal



→ difference between desired output and reference input.

→ In ideal case error should be zero.

$$C(s) = E(s) G(s)$$

$$E(s) = R(s) - C(s) H(s)$$

$$E(s) = R(s) - E(s) G(s) H(s)$$

$$E(s) + E(s) G(s) H(s) = R(s)$$

$$E(s) [1 + G(s) H(s)] = R(s)$$

$$E(s) = \frac{R(s)}{1 + G(s) H(s)}$$

in time domain:

$$e(t) = \mathcal{L}^{-1} \left[ \frac{R(s)}{1 + G(s) H(s)} \right]$$

steady state error:

$$e_{ss} = \lim_{t \rightarrow \infty} e(t)$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)H(s)}$$

unit step input:

$$R(s) = 1/s$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1/s}{1+G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1+G(s)H(s)}$$

$$= \frac{1}{1 + \lim_{s \rightarrow 0} G(s)H(s)}$$

$$e_{ss} = \frac{1}{1+k_p}$$

positional error constant:

$$k_p = \lim_{s \rightarrow 0} G(s)H(s)$$

type-0 system:

$$k_p = \lim_{s \rightarrow 0} K \frac{(s+z_1)(s+z_2)\dots}{(s+p_1)(s+p_2)\dots}$$

$$= K \frac{z_1 z_2 \dots}{p_1 p_2 \dots} = \text{Constant}$$

$$e_{ss} = \frac{1}{1+k_p} = \text{constant}$$

type-1 system:

$$k_p = \lim_{s \rightarrow 0} K \frac{(s+z_1)(s+z_2)\dots}{s(s+p_1)(s+p_2)\dots}$$

$$k_p = \infty$$

$$e_{ss} = \frac{1}{1+k_p} = \frac{1}{1+\infty} = 0$$

unit ramp input:

$$R(s) = 1/s^2$$

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{R(s)}{1 + G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} s \frac{1/s^2}{1 + G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s} \left[ \frac{1}{1 + G(s)H(s)} \right]$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + sG(s)H(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s + \lim_{s \rightarrow 0} sG(s)H(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} sG(s)H(s)}$$

$$e_{ss} = \frac{1}{K_v}$$

velocity error constant:  $K_v = \lim_{s \rightarrow 0} sG(s)H(s)$

type-0 system:

$$K_v = \lim_{s \rightarrow 0} s \cdot K \frac{(s+z_1)(s+z_2)\dots}{(s+p_1)(s+p_2)\dots}$$

$$K_v = 0$$

$$e_{ss} = 1/K_v = 1/0$$

$$e_{ss} = \infty$$

type-1 system:

$$K_V = \lim_{s \rightarrow 0} s \cdot K \frac{(s+z_1)(s+z_2)\dots}{s(s+p_1)(s+p_2)\dots}$$

$$= \lim_{s \rightarrow 0} K \frac{(s+z_1)(s+z_2)\dots}{(s+p_1)(s+p_2)\dots}$$

$$K_V = K \frac{z_1 z_2 \dots}{p_1 p_2 \dots} = \text{Constant}$$

$$e_{ss} = 1/K_V = \text{Constant}$$

type-2 system:

$$K_V = \lim_{s \rightarrow 0} s \cdot K \frac{(s+z_1)(s+z_2)\dots}{s^2(s+p_1)(s+p_2)\dots}$$

$$= \lim_{s \rightarrow 0} K \frac{(s+z_1)(s+z_2)\dots}{s(s+p_1)(s+p_2)\dots}$$

$$K_V = \infty$$

$$e_{ss} = 1/K_V = 1/\infty$$

$$e_{ss} = 0$$

unit parabolic input:

$$R(s) = 1/s^3$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{s (1/s^3)}{1 + G(s)H(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)H(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s^2 + \lim_{s \rightarrow 0} s^2 G(s)H(s)}$$

$$e_{ss} = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s) H(s)}$$

$$e_{ss} = 1/k_a$$

acceleration error constant:  $k_a = \lim_{s \rightarrow 0} s^2 G(s) H(s)$

type-0 system:

$$k_a = \lim_{s \rightarrow 0} s^2 K \frac{(s+z_1)(s+z_2) \dots}{(s+p_1)(s+p_2) \dots}$$

$$k_a = 0$$

$$e_{ss} = 1/0 = \infty$$

type-1 system:

$$k_a = \lim_{s \rightarrow 0} s^2 K \frac{(s+z_1)(s+z_2) \dots}{s(s+p_1)(s+p_2) \dots}$$

$$k_a = \lim_{s \rightarrow 0} s K \frac{(s+z_1)(s+z_2) \dots}{(s+p_1)(s+p_2) \dots}$$

$$k_a = 0$$

$$e_{ss} = \infty$$

type-2 system:

$$k_a = \lim_{s \rightarrow 0} s^2 K \frac{(s+z_1)(s+z_2) \dots}{s^2(s+p_1)(s+p_2) \dots}$$

$$k_a = \lim_{s \rightarrow 0} s \cdot \frac{(s+z_1)(s+z_2) \dots}{(s+p_1)(s+p_2) \dots}$$

$$= K \frac{z_1 z_2 \dots}{p_1 p_2 \dots}$$

$$k_a = \text{constant}$$

$$e_{ss} = 1/k_a = \text{constant}$$

type-3 system:

$$k_a = \lim_{s \rightarrow 0} s^2 \cdot \frac{(s+z_1)(s+z_2) \dots}{s^3 (s+p_1)(s+p_2) \dots}$$

$$k_a = \lim_{s \rightarrow 0} \frac{(s+z_1)(s+z_2) \dots}{s (s+p_1)(s+p_2) \dots}$$

$$k_a = \infty$$

$$e_{ss} = 1/\infty = 0$$

Static error coefficients:

→ positional error constant:  $K_p = \lim_{s \rightarrow 0} G(s) H(s)$

→ velocity error constant:  $K_v = \lim_{s \rightarrow 0} s G(s) H(s)$

→ acceleration error constant:  $K_a = \lim_{s \rightarrow 0} s^2 G(s) H(s)$

## Static error constants:

| error constant | type - 0 | type - 1 | type - 2 |
|----------------|----------|----------|----------|
| $K_p$          | Constant | $\infty$ | $\infty$ |
| $K_v$          | 0        | Constant | $\infty$ |
| $K_a$          | 0        | 0        | Constant |

## Steady state error:

| input signal   | type - 0          | type - 1        | type - 2        |
|----------------|-------------------|-----------------|-----------------|
| unit step      | $\frac{1}{1+K_p}$ | 0               | 0               |
| unit ramp      | $\infty$          | $\frac{1}{K_v}$ | 0               |
| unit parabolic | $\infty$          | $\infty$        | $\frac{1}{K_a}$ |

4. For servo mechanisms with open loop transfer function given below, explain what type of input signal give rise to a constant steady state error and calculate their values.

$$a) \quad G(s) = \frac{20(s+2)}{s(s+1)(s+3)}$$

$$\text{let } H(s) = 1$$

$$G(s)H(s) = \frac{20(s+2)}{s(s+1)(s+3)}$$

1 pole at origin  $\Rightarrow$  type-1 system

type-1 system  $\Rightarrow$  unit ramp input give a constant  $e_{ss}$

$$e_{ss} = \frac{1}{K_V}$$

$$K_V = \lim_{s \rightarrow 0} s G(s) H(s)$$

$$= \lim_{s \rightarrow 0} s G(s)$$

$$= \lim_{s \rightarrow 0} s \frac{20(s+2)}{s(s+1)(s+3)}$$

$$= \lim_{s \rightarrow 0} \frac{20(s+2)}{(s+1)(s+3)}$$

$$= \frac{20 \times 2}{1 \times 3}$$

$$K_V = \frac{40}{3}$$

$$e_{ss} = \frac{1}{k_v}$$

$$e_{ss} = \frac{3}{40} = 0.075$$

$$b) \quad G(s) = \frac{10}{(s+2)(s+3)}$$

$$\text{let } H(s) = 1$$

$$G(s) H(s) = \frac{10}{(s+2)(s+3)}$$

0 pole at origin  $\Rightarrow$  type-0 system

type-0 system  $\Rightarrow$  unit step input give a constant  $e_{ss}$

$$e_{ss} = \frac{1}{1+k_p}$$

$$k_p = \lim_{s \rightarrow 0} G(s) H(s)$$

$$= \lim_{s \rightarrow 0} G(s)$$

$$= \lim_{s \rightarrow 0} \frac{10}{(s+2)(s+3)}$$

$$= \frac{10}{2 \times 3}$$

$$k_p = \frac{5}{3}$$

$$e_{ss} = \frac{1}{1+k_p} = \frac{1}{1+5/3} = \frac{3}{8} = 0.375$$

$$c) G(s) = \frac{10}{s^2(s+1)(s+2)}$$

$$\text{Let } H(s) = 1$$

$$G(s)H(s) = \frac{10}{s^2(s+1)(s+2)}$$

2 poles at origin  $\Rightarrow$  type-2 system

type-2 system  $\Rightarrow$  unit parabolic input give a constant  $e_{ss}$

$$e_{ss} = \frac{1}{k_a}$$

$$k_a = \lim_{s \rightarrow 0} s^2 G(s) H(s)$$

$$= \lim_{s \rightarrow 0} s^2 G(s)$$

$$= \lim_{s \rightarrow 0} s^2 \frac{10}{s^2(s+1)(s+2)}$$

$$= \lim_{s \rightarrow 0} \frac{10}{(s+1)(s+2)}$$

$$k_a = \frac{10}{2} = 5$$

$$e_{ss} = \frac{1}{k_a} = \frac{1}{5}$$

$$e_{ss} = 0.2$$

5. For a unity feedback control system The open loop transfer function is given below. Find (a) the position, velocity and acceleration error constants. b) the steady state error when the input is  $R(s)$ .

$$G(s) = \frac{10(s+2)}{s^2(s+1)}$$

$$R(s) = \frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}$$

unity feed back system:

$$H(s) = 1$$

$$k_p = \lim_{s \rightarrow 0} G(s)H(s)$$

$$= \lim_{s \rightarrow 0} G(s)$$

$$= \lim_{s \rightarrow 0} \frac{10(s+2)}{s^2(s+1)}$$

position error constant:  $k_p = \infty$

$$k_v = \lim_{s \rightarrow 0} s G(s)H(s)$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{10(s+2)}{s^2(s+1)}$$

$$= \lim_{s \rightarrow 0} \frac{10(s+2)}{s(s+1)}$$

velocity error constant:  $k_v = \infty$

$$k_a = \lim_{s \rightarrow 0} s^2 G(s)H(s)$$

$$= \lim_{s \rightarrow 0} s^2 \cdot \frac{10(s+2)}{s^2(s+1)}$$

$$= \lim_{s \rightarrow 0} \frac{10(s+2)}{(s+1)}$$

$$= \frac{10(2)}{1}$$

acceleration error constant:  $K_a = 20$

error signal:

$$E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

$$= \frac{\frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}}{1 + \frac{10(s+2)}{s^2(s+1)}}$$

$$= \frac{\frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}}{\frac{s^2(s+1) + 10(s+2)}{s^2(s+1)}}$$

$$E(s) = \frac{3}{s} \left[ \frac{s^2(s+1)}{s^2(s+1) + 10(s+2)} \right] - \frac{2}{s^2} \left[ \frac{s^2(s+1)}{s^2(s+1) + 10(s+2)} \right] + \frac{1}{3s^3} \left[ \frac{s^2(s+1)}{s^2(s+1) + 10(s+2)} \right]$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} s \left\{ \frac{3}{s} \left[ \frac{s^2(s+1)}{s^2(s+1) + 10(s+2)} \right] - \frac{2}{s^2} \left[ \frac{s^2(s+1)}{s^2(s+1) + 10(s+2)} \right] + \frac{1}{3s^3} \left[ \frac{s^2(s+1)}{s^2(s+1) + 10(s+2)} \right] \right\}$$

$$= \lim_{s \rightarrow 0} \left\{ 3 \left[ \frac{s^2(s+1)}{s^2(s+1) + 10(s+2)} \right] - 2 \left[ \frac{s(s+1)}{s^2(s+1) + 10(s+2)} \right] + \frac{1}{3} \left[ \frac{(s+1)}{s^2(s+1) + 10(s+2)} \right] \right\}$$

$$= [0 - 0 + 1/60]$$

$$e_{ss} = 1/60$$

6. Consider a unity feedback system with a closed loop transfer function is given below. Determine open loop transfer Function  $G(s)$ . Show that steady state error with unit ramp input is given by  $(a-K) / b$ .

$$\frac{C(s)}{R(s)} = \frac{ks+b}{s^2+as+b}$$

unity feedback system:

$$H(s) = 1$$

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$T(s) = \frac{G(s)}{1+G(s)}$$

$$T(s) + T(s)G(s) = G(s)$$

$$T(s) = G(s) - T(s)G(s)$$

$$T(s) = G(s) [1 - T(s)]$$

$$G(s) = \frac{T(s)}{1 - T(s)}$$

$$= \frac{ks+b / s^2+as+b}{1 - \frac{ks+b}{s^2+as+b}}$$

$$= \frac{ks+b}{s^2+as+b-ks-b}$$

$$G(s) = \frac{ks+b}{s^2+(a-k)s}$$

unit ramp input  $\Rightarrow$  velocity error constant

$$\begin{aligned}K_V &= \lim_{s \rightarrow 0} s G(s) H(s) \\&= \lim_{s \rightarrow 0} s \cdot \frac{ks+b}{s^2+(a-k)s} \\&= \lim_{s \rightarrow 0} s \cdot \frac{ks+b}{s[s+(a-k)]} \\&= \lim_{s \rightarrow 0} \frac{ks+b}{s+a-k} \\K_V &= \frac{b}{a-k}\end{aligned}$$

steady state error:

$$\begin{aligned}e_{ss} &= \frac{1}{K_V} \\e_{ss} &= \left( \frac{a-k}{b} \right)\end{aligned}$$

7. The open loop transfer function of a servo system with unity feedback is given below. Evaluate the static error constants of the system. obtain the steady state error of the system, when subjected to an input given by the polynomial  $r(t)$ .

$$G(s) = \frac{10}{s(0.1s+1)}$$

$$r(t) = a_0 + a_1 t + \frac{a_2 t^2}{2}$$

unity feedback:

$$H(s) = 1$$

loop transfer function:

$$G(s)H(s) = \frac{10}{s(0.1s+1)} \times 1$$

$$G(s)H(s) = \frac{10}{s(0.1s+1)}$$

position error constant:

$$K_p = \lim_{s \rightarrow 0} G(s)H(s)$$

$$= \lim_{s \rightarrow 0} \frac{10}{s(0.1s+1)}$$

$$K_p = \infty$$

velocity error constant:

$$K_v = \lim_{s \rightarrow 0} s G(s)H(s)$$

$$= \lim_{s \rightarrow 0} s \frac{10}{s(0.1s+1)}$$

$$= \frac{10}{1}$$

$$K_v = 10$$

acceleration error constant:

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s)$$

$$= \lim_{s \rightarrow 0} s^2 \frac{10}{s(0.1s+1)}$$

$$K_a = 0$$

error signal:

$$E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

$$R(s) = \frac{a_0}{s} + \frac{a_1}{s^2} + \frac{a_2}{s^3}$$

$$E(s) = \frac{\frac{a_0}{s} + \frac{a_1}{s^2} + \frac{a_2}{s^3}}{1 + \frac{10}{s(0.18s+1)}}$$

$$= \frac{\frac{a_0}{s} + \frac{a_1}{s^2} + \frac{a_2}{s^3}}{\frac{s(0.18s+1) + 10}{s(0.18s+1)}}$$

$$= \left[ \frac{a_0}{s} + \frac{a_1}{s^2} + \frac{a_2}{s^3} \right] * \left[ \frac{s(0.18s+1)}{s(0.18s+1) + 10} \right]$$

$$E(s) = \frac{a_0}{s} \left[ \frac{s(0.18s+1)}{s(0.18s+1)+10} \right] + \frac{a_1}{s^2} \left[ \frac{s(0.18s+1)}{s(0.18s+1)+10} \right] + \frac{a_2}{s^3} \left[ \frac{s(0.18s+1)}{s(0.18s+1)+10} \right]$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} s \left\{ \frac{a_0}{s} \left[ \frac{s(0.18s+1)}{s(0.18s+1)+10} \right] + \frac{a_1}{s^2} \left[ \frac{s(0.18s+1)}{s(0.18s+1)+10} \right] + \frac{a_2}{s^3} \left[ \frac{s(0.18s+1)}{s(0.18s+1)+10} \right] \right\}$$

$$= \lim_{s \rightarrow 0} a_0 \left[ \frac{s(0.18s+1)}{s(0.18s+1)+10} \right] + a_1 \left[ \frac{(0.18s+1)}{s(0.18s+1)+10} \right] + \frac{a_2}{s} \left[ \frac{(0.18s+1)}{s(0.18s+1)+10} \right]$$

$$= a_0(0) + a_1 \left[ \frac{1}{10} \right] + \infty$$

$$e_{ss} = \infty$$